

STAT-4320 Notes

June 29, 2026

Contents

1 Class 1	5
1.1 The sample mean	5
1.2 Central Limit Theorem	5
2 Chapter 2	6
2.1 Breakdown Point and Efficiency	6
2.1.1 Sample and population median	6
2.1.2 Breakdown point	6
2.1.3 Efficiency	6
3 Class 3	8
3.1 Convergence of random variables	8
3.2 Slutsky's Lemma and Continuous Mapping Theorem	8
3.3 Weak Law of Large Numbers and Central Limit Theorem	8
3.4 Delta method	9
3.5 Multivariate Data	10
3.6 Multivariate Normal	11
4 Class 4	12
4.1 Linear Algebra Review	12
4.1.1 Positive definite matrices	12
4.2 Moment generating functions	13
4.3 Properties of the multivariate normal	13
4.4 Sample Variance	14
5 Class 5	15
5.1 Chi-squared distribution	15
5.2 Sample Variance, cont'd	15
5.3 Joint distribution of sample mean and sample variance	17
6 Class 6	18
6.1 Basic Framework of Statistical Estimation	18
6.2 Properties of Estimators	18
7 Class 7	22
7.1 Consistency	22
7.2 Method of moments estimation	22
8 Class 8	26
8.1 Maximum Likelihood Estimation	26
9 Class 9	29
9.1 Fisher Information	29
9.2 Asymptotic properties of the MLE estimator	29
9.3 Asymptotic properties of the MLE estimator for multiple parameters	32
10 Class 10	34
10.1 Cramer-Rao Lower Bound	34
10.2 Simple Linear Regression	35
10.2.1 Estimating β_0, β_1	35
10.2.2 Estimating variance	36
10.2.3 Facts about Least Square Estimates	36
11 Class 11	38
11.1 Least Absolute Deviation Line	38
12 Class 12	40
12.1 Confidence Intervals	40
12.1.1 Exact Confidence Intervals	40
12.1.2 Asymptotic Confidence Intervals	41
13 Class 13	43
13.1 Proof of Weak of Law of Large Numbers without finite variance	43

14 Class 14	44
14.1 Hypothesis testing	44
14.1.1 Likelihood Ratio Test and Neyman-Pearson Lemma	44
14.1.2 One sided composite hypothesis and UMP tests	46
15 Class 15	47
15.1 Generalized Likelihood Ratio Test	47
15.2 Asymptotic distribution of the GLRT	48
15.3 GLRT for submodels	48
16 Lecture 16	50
16.1 GLRT for submodels, cont'd	50
17 Class 17	53
17.1 Multiple linear regression	53
17.1.1 Setup	53
17.1.2 Least Square Estimate	53
17.1.3 Properties of the LS estimate	54
17.1.4 Geometric interpretations of LS estimates	54
18 Class 18	56
18.1 Distribution of residuals	56
18.1.1 Estimating σ^2 in MLR	56
18.2 Confidence interval and hypothesis testing for regression coefficients	57
18.2.1 F-test and ANOVA	58
19 Class 19	60
19.1 Comparison of confidence intervals	60
19.2 Confidence set	60
19.3 Hypothesis tests	61
19.3.1 Simple vs simple	61
19.3.2 Simple vs composite	61
19.3.3 Composite vs composite	62
19.3.4 Asymptotic power of tests	62
20 Lecture 22	64
20.1 Multicollinearity	64
20.2 Variance Inflation Factor (VIF)	64
20.3 Interpretation of VIF	64
20.4 Methods to Handle Multicollinearity	64
20.5 Ridge Regression	64
20.6 Centering and Scaling	65
20.7 Constrained Form of Ridge Regression	65
20.8 Matrix Form	65
21 Lecture 23	66
21.1 Properties of the Ridge Regression Estimator	66
21.2 Mean-Square Error for a Multivariate Estimator	66
21.3 Degrees of Freedom in Ridge Regression	66
21.4 Selecting λ : Generalized Cross-Validation (GCV)	67
21.5 Alternate Constrained Form of Ridge Regression	67
21.6 Best Subset Selection	67
21.7 LASSO (Least Absolute Shrinkage and Selection Operator)	67
22 Lecture 24	68
22.1 Best Subset Selection	68
22.2 LASSO (Least Absolute Shrinkage and Selection Operator)	68
22.2.1 Lagrangian Form	68
22.3 Closed Form Shrinkage When $X = I$	68
22.4 Selecting λ in LASSO: K -Fold Cross-Validation	68
22.5 Elastic Net	69
22.6 Logistic Regression	69
22.7 Likelihood and Estimation	69
22.8 GLRT for Testing a Coefficient	69

22.9 Confidence Interval for β_j	69
23 Appendix	70

1 Class 1

1.1 The sample mean

Definition 1.1. (Sample mean): Given $X_1, \dots, X_n$ iid from F , and $\mathbb{E}[X_i] = \mu$, the sample mean is the random variable defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

Remark. The sample mean is random, so it has an expectation

$$\begin{aligned} \mathbb{E}[\bar{X}] &= \mathbb{E}\left[\frac{1}{n} \sum_{i=1}^n X_i\right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{E}[X_i] \quad \text{by linearity} \\ &= \mu \end{aligned}$$

The expectation of the sample mean is the population mean.

The sample mean is an unbiased estimator of the population mean.

Remark.

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{1}{n} \sum_{i=1}^n X_i\right) \\ &= \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) \quad \text{since } \mathbb{E}[cX_i] = c\mathbb{E}[X_i], \text{Var}(cX_i) = c^2\text{Var}(X_i), X_i \text{ independent} \\ &= \frac{\sigma^2}{n} \end{aligned}$$

Remark. We say that $\bar{X}$ follows a sampling distribution. Think of this as a thought experiment. If a sample of size n is taken many times, we expect to see the sample mean exhibit the above expectation and variability.

1.2 Central Limit Theorem

Theorem 1.2. Theorem (Central Limit Theorem): Let $X_1, \dots, X_n$ be iid from arbitrary distribution F with mean μ and variance σ^2 , then as $n \rightarrow \infty$,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0, 1),$$

OR

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

2 Chapter 2

2.1 Breakdown Point and Efficiency

Recall that sample mean is defined as

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

An issues with the sample mean is that corruption in 1 or few data points can make sample mean unstable. The sample median is more robust alternative.

2.1.1 Sample and population median

Definition 2.1. (Sample median): The sample median is the *middle value* when a list of numbers are sorted in non-decreasing order.

$$X_{med} = \begin{cases} X_{(\frac{n+1}{2})} & \text{if } n \text{ odd} \\ X_{(\frac{n}{2})} + X_{(\frac{n}{2}+1)} & \text{if } n \text{ even} \end{cases}$$

Where $X_{(i)}$ denotes the i -th smallest value in $X_1, \dots, X_n$

Definition 2.2. (population median): The population median of distribution F with density function f is the point m such that

$$\int_{-\infty}^m f(x)dx = \int_m^{\infty} f(x)dx = \frac{1}{2}$$

2.1.2 Breakdown point

Definition 2.3. (Breakdown Point): The breakdown point of an estimate $\hat{\theta}_n$ based on data $X_1 \dots X_n$ is the fraction of data points that have to be moved to infinity for the estimate to also move to infinity.

Example. The breakdown point

- For sample mean $= \frac{1}{n}$
- For sample median $\approx \frac{1}{2}$

Remark. Note that this is **not** a direct consequence of CLT.

For example, $F = N(\mu, \sigma^2)$.

The sample mean follows **exactly** a normal distribution

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

The sample median approximately follows

$$X_{med} \approx N\left(\mu, \frac{1}{4f(\mu)^2 n}\right)$$

Recall that

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2} \frac{(x-\mu)^2}{\sigma^2}}$$

Hence

$$f(\mu) = \frac{1}{\sqrt{2\pi}\sigma}$$
$$X_{med} \approx N\left(\mu, \frac{\pi\sigma^2}{2n}\right)$$

2.1.3 Efficiency

Definition 2.4. (Efficiency): The efficiency of two estimates is the ratio of their variances.

$$\text{Efficiency} \left(\tilde{X}_{med}, \bar{X} \right) = \frac{\text{Var}(\bar{X})}{\text{Var}(\tilde{X}_{med})}$$

Example. For sample mean and sample median, the efficiency is $\frac{2}{\pi}$.

3 Class 3

3.1 Convergence of random variables

There are two kinds of convergence

- convergence in probability
- convergence in distribution

Definition 3.1. (Convergence in probability): We say a sequence of random variables $\{X_n\}_{n \geq 1}$ converges in probability to X if for any $\epsilon > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|X_n - X| > \epsilon) = 0$$

Denoted

$$X_n \xrightarrow{p} X$$

Definition 3.2. (Convergence in distribution): We say a sequence of random variables $\{X_n\}_{n \geq 1}$ converges in distribution to X if

$$\mathbb{P}(X_n \leq x) \xrightarrow{n \rightarrow \infty} \mathbb{P}(X \leq x)$$

Which is equivalent to

$$F_n(x) \xrightarrow{n \rightarrow \infty} F(x)$$

Denoted

$$X_n \xrightarrow{d} X$$

3.2 Slutsky's Lemma and Continuous Mapping Theorem

Lemma 3.3. Lemma (Slutsky's Lemma): If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{p} c$ for constant c , then the following hold

- $X_n + Y_n \xrightarrow{d} X + c$
- $X_n Y_n \xrightarrow{d} cX$
- $\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}$ if $c > 0$

Theorem 3.4. (Continuous mapping): If g is a continuous function, then

$$X_n \xrightarrow{p} X \implies g(X_n) \xrightarrow{p} g(X)$$

$$X_n \xrightarrow{d} X \implies g(X_n) \xrightarrow{d} g(X)$$

3.3 Weak Law of Large Numbers and Central Limit Theorem

The weak law of large numbers is an example of convergence in probability. The central limit theorem is an example of convergence in distribution.

Theorem 3.5. Weak law of large numbers: Suppose $X_1 \dots X_n$ iid from F with $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X) = \sigma^2$, then

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mu$$

Proof.

$$\begin{aligned} \mathbb{P}(|\bar{X} - \mu| > \epsilon) &= \frac{\text{Var}(\bar{X})}{\epsilon^2} \quad \text{by Chebyshev's Inequality} \\ &= \frac{\sigma^2}{n\epsilon^2} \\ \frac{\sigma^2}{n\epsilon^2} &\xrightarrow{n \rightarrow \infty} 0 \end{aligned}$$

✓

Theorem 3.6. Markov's Inequality: For any random variable X , and non-negative constant a ,

$$P(|X| \geq a) \leq \frac{\mathbb{E}[|X|]}{a}$$

Alternatively, for any non-negative random variable X ,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}$$

Proof. We prove the general case, for any random variable X , let $Y = |X|$

$$\begin{aligned} \mathbb{E}[Y] &= \mathbb{E}[Y|Y \geq a] P(Y \geq a) + \mathbb{E}[Y|Y < a] P(Y < a) \text{ by Law of Total Expectation} \\ &\geq \mathbb{E}[Y|Y \geq a] P(Y \geq a) \\ &\geq a P(Y \geq a) \\ \implies P(Y \geq a) &\leq \frac{\mathbb{E}[Y]}{a} \end{aligned}$$

✓

Theorem 3.7. Chebyshev's Inequality: Let X be a random variable with mean μ and variance σ^2 , then for any $a > 0$

$$P(|X - \mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

Proof.

$$\begin{aligned} P(|X - \mu| \geq a) &= P((X - \mu)^2 \geq a^2) \\ &\leq \frac{\mathbb{E}[(X - \mu)^2]}{a^2} \quad \text{by Markov's Inequality} \\ &= \frac{\text{Var}(X)}{a^2} \end{aligned}$$

✓

Theorem 3.8. Central limit theorem: Suppose $X_1 \dots X_n$ iid from F with $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X) = \sigma^2$, then

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

Alternatively, let

$$\begin{aligned} Z_n &= \frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \\ \mathbb{P}(Z_n \leq t) &= P(N(0, 1) \leq t) \text{ for all } t \in \mathbb{R} \end{aligned}$$

Remark.

$$\bar{X}_n \approx N\left(\mu, \frac{\sigma^2}{n}\right)$$

Although this statement has no mathematical content. We are taking the limit of the distribution of $\bar{X}$ as n gets large. $N\left(\mu, \frac{\sigma^2}{n}\right)$ cannot be a limit.

3.4 Delta method

CLT gives asymptotic distribution of $\bar{X}$. We want to get the asymptotic distribution of functions of $\bar{X}$. By Continuous Mapping Theorem, we get this for free

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1) \implies g\left(\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma}\right) \xrightarrow{d} g(N(0, 1))$$

However, we don't just want statements about $g(Z_n)$, we want statements about $g(\bar{X})$

Theorem 3.9. Delta Method: Suppose $X_1, \dots, X_n$ iid F , with $\mathbb{E}[X_1] = \mu$ and $\text{Var}(X) = \sigma^2$ and g is a function

such that the derivative of $g'(\mu) \neq 0$. Then

$$\sqrt{n} (g(\bar{X}) - g(\mu)) \xrightarrow{d} N(0, \sigma^2 (g'(\mu))^2)$$

Remark. Note: We know by Continuous Mapping Theorem that the in-probability limit of $g(\bar{X})$ is $g(\mu)$

$$g(\bar{X}) \xrightarrow{p} g(\mu)$$

Subtracting away the in-probability limit and taking the Z-score, delta method tells us that the z-score follows a normal distribution.

Proof.

Recall Taylor's Expansion

$$f(x) = f(a) + (x - a)f'(a) + \frac{1}{2}(x - a)^2 f''(a) + \dots$$

$$g(\bar{X}) - g(\mu) = (\bar{X} - \mu)g'(\mu) + \text{error terms}$$

$$\sqrt{n} (g(\bar{X}) - g(\mu)) = \sqrt{n}(\bar{X} - \mu)g'(\mu) + \text{error terms}$$

By CLT, we know that

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

By Slutsky's,

$$\sqrt{n}(\bar{X} - \mu)g'(\mu) \xrightarrow{d} N(0, \sigma^2)g'(\mu) = N(0, \sigma^2 (g'(\mu))^2)$$

Note: if $g'(\mu) = 0$, by taking higher orders in the Taylor expansion, we get

$$g(\bar{X}) - g(\mu) \approx \frac{1}{2} (\bar{X} - \mu)^2 g''(\mu)$$

Since

$$n (\bar{X} - \mu)^2 = (\sqrt{n} (\bar{X} - \mu))^2 \xrightarrow{d} (\sigma N(0, 1))^2 = \sigma^2 \chi_1^2$$

We get

$$n (g(\bar{X}) - g(\mu)) \xrightarrow{d} \frac{1}{2} \sigma^2 \chi_1^2 g''(\mu)$$

✓

3.5 Multivariate Data

For each unit of study, the number of measurements is greater than 1. For example, for n data points and p observed variables

$$\mathbf{X}_1 = \begin{pmatrix} X_{11} \\ X_{12} \\ \vdots \\ X_{1p} \end{pmatrix}, \mathbf{X}_2 = \begin{pmatrix} X_{21} \\ X_{22} \\ \vdots \\ X_{2p} \end{pmatrix}, \mathbf{X}_3 = \begin{pmatrix} X_{31} \\ X_{32} \\ \vdots \\ X_{3p} \end{pmatrix}$$

Where $\mathbf{X}_i$ are iid p -dimensional observations with distribution F .

The mean vector is

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}_1] = \begin{pmatrix} \mathbb{E}[X_{11}] \\ \mathbb{E}[X_{12}] \\ \vdots \\ \mathbb{E}[X_{1p}] \end{pmatrix} \in \mathbb{R}^p$$

The covariance matrix is denoted

$$\boldsymbol{\Sigma} \in \mathbb{R}^{p \times p}$$

Where the $i - j$ -th element is

$$\boldsymbol{\Sigma}_{i,j} = \text{Cov}(X_{1i}, X_{1j}) = \mathbb{E}[X_{1i}X_{1j}] - \mathbb{E}[X_{1i}] \mathbb{E}[X_{1j}]$$

Hence, we can express $\boldsymbol{\Sigma}$ as the difference of two matrices

$$\boldsymbol{\Sigma} = \mathbb{E}[\mathbf{X}_1 \mathbf{X}_1^T] - \mathbb{E}[\mathbf{X}] \mathbb{E}[\mathbf{X}]^T$$

3.6 Multivariate Normal

Definition 3.10. (Multivariate Normal): A p -dimensional random vector $\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$ is said to follow the multivariate normal distribution with mean vector $\boldsymbol{\mu}$ and positive definite (pd) covariance matrix $\boldsymbol{\Sigma}$ if it has a density function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ of the form

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^p \det(\boldsymbol{\Sigma})}} e^{-\frac{1}{2}(\mathbf{x}-\boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x}-\boldsymbol{\mu})}$$

Remark. This definition only works with a pd covariance matrix.

Remark. A multivariate normal can be defined without the pd matrix, using the linear combination definition.

Definition 3.11. (Multivariate CLT): Suppose $\mathbf{X}_1 \dots \mathbf{X}_n$ are iid p -dimensional random vectors with mean vector $\boldsymbol{\mu} \in \mathbb{R}^p$ and covariance matrix $\boldsymbol{\Sigma}$
The sample mean

$$\bar{\mathbf{X}} = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i$$

has the following distribution

$$\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu}) \xrightarrow{d} N_p(\mathbf{0}, \boldsymbol{\Sigma})$$

4 Class 4

4.1 Linear Algebra Review

4.1.1 Positive definite matrices

Definition 4.1. Definition (positive definite): A symmetric $p \times p$ matrix is said to be positive definite (pd) if for all $\mathbf{x} \in \mathbb{R}^p \setminus \{0\}$,

$$\mathbf{x}^T A \mathbf{x} > 0$$

Remark. All eigenvalues of a pd matrix are positive

Remark. By spectral decomposition, any pd matrix can be written as

$$A = P \Lambda P^T,$$

Where Λ is a diagonal matrix with the eigenvalues of A on the diagonals

$$\Lambda = \begin{bmatrix} \lambda_1 & \dots & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ & \vdots & & \\ 0 & \dots & \dots & \lambda_p \end{bmatrix}$$

and P is an orthogonal matrix

$$P^T P = P P^T = I_{p \times p}$$

Remark. A^{-1} exists and is given by

$$A^{-1} = P \Lambda^{-1} P^T$$

Proof.

$$A^{-1} A = P \Lambda^{-1} P^T P \Lambda P^T = P \Lambda^{-1} \Lambda P^T = P P^T = I$$

✓

Remark. A has a square root. Given a pd matrix A , we say that B is the square root of A if $BB = A$

$$B = P \begin{bmatrix} \sqrt{\lambda_1} & \dots & \dots & 0 \\ 0 & \sqrt{\lambda_2} & \dots & 0 \\ & \vdots & & \\ 0 & \dots & \dots & \sqrt{\lambda_p} \end{bmatrix} P^T$$

Remark. Sum of eigenvalues is the trace of A

$$\sum_{i=1}^n \lambda_i = \text{tr}(A)$$

Remark. What does it mean to assume that the covariance matrix is pd ?

Consider any $\mathbf{a} \in \mathbb{R}^p \setminus \{0\}$. Recall that

$$\begin{aligned} \Sigma &= \mathbb{E} [\mathbf{X} \mathbf{X}^T] - \mathbb{E} [\mathbf{X}] \mathbb{E} [\mathbf{X}]^T \\ &= \mathbb{E} [(\mathbf{X} - \mathbb{E} [\mathbf{X}])(\mathbf{X} - \mathbb{E} [\mathbf{X}])^T] \end{aligned}$$

Consider $\mathbf{a}^T \Sigma \mathbf{a}$,

$$\begin{aligned} \mathbf{a}^T \Sigma \mathbf{a} &= \mathbb{E} [(\mathbf{a}^T (\mathbf{X} - \mathbb{E} [\mathbf{X}])(\mathbf{a}^T (\mathbf{X} - \mathbb{E} [\mathbf{X}]))^T] \text{ since } (AB)^T = B^T A^T \\ &= \mathbb{E} [(\mathbf{a}^T (\mathbf{X} - \mathbb{E} [\mathbf{X}]))^2] \\ &= \text{Var}(\mathbf{a}^T \mathbf{X}) \\ &> 0 \end{aligned}$$

i.e. *projected onto any direction $\mathbf{a}$, the variance of $\mathbf{X}$ is nonzero.*

i.e. The RV is non-degenerate along every direction $\mathbf{a} \in \mathbb{R}^p \setminus \{0\}$.

4.2 Moment generating functions

Definition 4.2. (Moment generating function) For a random variable X , the *mgf* is a function $\mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$,

$$\phi_X(t) = \mathbb{E}[e^{tX}]$$

Example. For $X \sim N(\mu, \sigma^2)$

$$\phi_X(t) = \exp\left(t\mu + \frac{1}{2}\sigma^2 t^2\right)$$

If X is a p -dimensional random variable, the *mgf* is a function $\mathbb{R}^p \rightarrow \mathbb{R}_{\geq 0}$

$$\phi_X(\mathbf{t}) = \exp\left(\mathbf{t}^T \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^T \boldsymbol{\Sigma} \mathbf{t}\right)$$

4.3 Properties of the multivariate normal

Proposition 4.3. If $\mathbf{X}$ is a p -dimensional normal, $\mathbf{X} \sim N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and $\mathbf{A}$ is a $k \times p$ matrix such that $\text{rank}(\mathbf{A}) = k \leq p$ (i.e. full row rank) and $\mathbf{b} \in \mathbb{R}^k$ is a fixed vector,

$$\mathbf{A}\mathbf{X} + \mathbf{b} \sim N_k(\mathbf{A}\boldsymbol{\mu}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^T)$$

Example. An important case of this is when $\mathbf{b} = \mathbf{0}$ and $k = 1$, then $\mathbf{A}$ is a row vector $\mathbf{A} = [a_1, a_2, \dots, a_p]$. Take any $\mathbf{a} \in \mathbb{R}^p$, then

$$\mathbf{a}^T \mathbf{X} \sim N_1(\mathbf{a}^T \boldsymbol{\mu}, \mathbf{a}^T \boldsymbol{\Sigma} \mathbf{a})$$

This can also be expressed as

$$\mathbf{a}^T \mathbf{X} = \sum_{i=1}^p a_i X_i \sim N_1\left(\sum_{i=1}^p a_i \mu_i, \mathbf{a}^T \boldsymbol{\Sigma} \mathbf{a}\right)$$

Proof. **Exercise for enthusiasts:** Prove 1 using *mgfs*. ✓

Proposition 4.4. Suppose $\mathbf{X}$ is a p_1 -dimensional RV and $\mathbf{Y}$ is a p_2 -dimensional RV, such that

$$\begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} \in \mathbb{R}^{p_1+p_2}, \begin{pmatrix} \mathbf{X} \\ \mathbf{Y} \end{pmatrix} \sim N_{p_1+p_2}\left(\begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}\right)$$

where $\boldsymbol{\mu}_1 \in \mathbb{R}^{p_1}$, $\boldsymbol{\mu}_2 \in \mathbb{R}^{p_2}$, and

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix} \in \mathbb{R}^{(p_1+p_2) \times (p_1+p_2)}$$

then we say that

$$\mathbf{X} \perp \mathbf{Y} \Leftrightarrow \boldsymbol{\Sigma}_{12} = \boldsymbol{\Sigma}_{21}^T = \mathbf{0}$$

Proof. The forward direction is obvious.

The converse is not true in general for a 1-dimensional normal, but it is true in multivariate normal. ✓

Example. Important case: $p_1 = p_2 = 1$, Suppose

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N\left((\mu_1, \mu_2), \begin{bmatrix} \sigma_1^2 & \sigma_{21} \\ \sigma_{12}^2 & \sigma_2^2 \end{bmatrix}\right)$$

Then $X \perp Y$ if and only if $\sigma_{12} = 0$.

Remark. Relationship

- If $X \perp Y$, then $\text{Cov}(X, Y) = 0$. This is always true

$$X \perp Y \implies \text{Cov}(X, Y) = 0$$

- The converse is not true in general
- However, if (X, Y) are jointly normal or jointly bernoulli, then

$$\text{Cov}(X, Y) = 0 \implies X \perp Y$$

4.4 Sample Variance

The sample variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Has a sampling distribution

$$s^2 \sim \frac{\sigma^2}{n-1} \chi_{n-1}^2$$

5 Class 5

5.1 Chi-squared distribution

Definition 5.1. The χ_d^2 is the sum of d iid standard normals squared.

$$Z_1, \dots, Z_d \sim N(0, 1), \quad \sum_{i=1}^d Z_i^2 \sim \chi_d^2$$

Result 5.2. The *chi-squared* distribution has expectation

$$\begin{aligned} \mathbb{E} [\chi_d^2] &= d\mathbb{E} [Z_1^2] \\ &= d \end{aligned}$$

Proof.

$$\mathbb{E} [Z_1^2] = \text{Var}(Z_1) + \mathbb{E} [Z_1]^2$$

✓

Result 5.3. The *chi-squared* distribution has variance

$$\begin{aligned} \text{Var}(\chi_d^2) &= \text{Var} \left(\sum_{i=1}^d Z_i^2 \right) \\ &= \sum_{i=1}^d \text{Var}(Z_i^2) \\ &= 2d \end{aligned}$$

Proof.

$$\begin{aligned} \text{Var}(Z_1^2) &= \mathbb{E} [Z_1^4] - \mathbb{E} [Z_1^2]^2 \\ &= \mathbb{E} [Z_1^4] - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

✓

5.2 Sample Variance, cont'd

Result 5.4. The sample variance follows a *chi-squared* distribution.

$$s^2 \sim \frac{\sigma^2}{n-1} \chi_{n-1}^2$$

Proof. Let

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} = \begin{pmatrix} X_1 - \bar{X} \\ X_2 - \bar{X} \\ \vdots \\ X_n - \bar{X} \end{pmatrix}$$

We express $\mathbf{Y} = \mathbf{A}\mathbf{X}$ for some matrix $\mathbf{A}$.

$$\mathbf{Y} = \begin{pmatrix} 1 - \frac{1}{n} & -\frac{1}{n} & \cdots & -\frac{1}{n} \\ -\frac{1}{n} & 1 - \frac{1}{n} & \cdots & -\frac{1}{n} \\ \vdots & \vdots & \ddots & \vdots \\ -\frac{1}{n} & -\frac{1}{n} & \cdots & 1 - \frac{1}{n} \end{pmatrix} \mathbf{X}$$

Since

$$\mathbf{X} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$$

Hence

$$\begin{aligned}
\mathbf{Y} &\sim N_n(\mathbf{A}\boldsymbol{\mu}, \sigma^2 \mathbf{A}\mathbf{A}^T) \\
&= N_n(\mu \mathbf{A}\mathbf{1}, \sigma^2 \mathbf{A}\mathbf{A}^T) \\
&= N_n(\mathbf{0}, \sigma^2 \mathbf{A}\mathbf{A}^T) \text{ since } \mathbf{A}\mathbf{1} = \mathbf{0} \\
&= N_n(\mathbf{0}, \sigma^2 \mathbf{A}^2) \text{ since } \mathbf{A} \text{ symmetric}
\end{aligned}$$

Note that

$$\mathbf{A} = \mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^T$$

Where $\mathbf{1}\mathbf{1}^T$ is a rank 1 symmetric matrix. Hence $\mathbf{1}\mathbf{1}^T$ has at most 1 non-zero eigenvalue. Since the sum of eigenvalues is the trace, and $\text{tr}(\mathbf{1}\mathbf{1}^T) = n$, the non-zero eigenvalue is n . $\mathbf{1}\mathbf{1}^T$ has eigenvalues $(n, 0, 0, \dots, 0)$. $\frac{1}{n} \mathbf{1}\mathbf{1}^T$ has eigenvalues $(1, 0, 0, \dots)$.

Therefore, $\mathbf{I} - \frac{1}{n} \mathbf{1}\mathbf{1}^T$ has eigenvalues $(1, 1, \dots, 1, 0)$ (This only works because of $\mathbf{I}$).

We can express s^2 in terms of $\mathbf{Y}$.

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n-1} \mathbf{Y}^T \mathbf{Y}$$

It suffices to show that

$$\mathbf{Y}^T \mathbf{Y} \sim \sigma^2 \chi_{n-1}^2$$

Note: $\mathbf{Y}$ has elements that are normal, i.e. $Y_i = X_i - \bar{X}$ is a difference of normals. However, Y_i is not independent due to $\bar{X}$.

Fact: Denote $\boldsymbol{\Sigma} = \mathbf{A}^2$, and we define $\mathbf{Z} = N_n(\mathbf{0}, \mathbf{I})$. Then,

$$\mathbf{Y} \stackrel{d}{=} \sigma \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{Z} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{A})$$

This is true because

$$\begin{aligned}
\sigma \boldsymbol{\Sigma}^T \boldsymbol{\Sigma} &\sim N_n\left(\mathbf{0}, \sigma^2 \left(\boldsymbol{\Sigma}^{\frac{1}{2}}\right)^T \boldsymbol{\Sigma}^{\frac{1}{2}}\right) \\
&= N_n(\mathbf{0}, \sigma^2 \mathbf{A}^T \mathbf{A}) \\
&= N_n(\mathbf{0}, \sigma^2 \mathbf{A}^2)
\end{aligned}$$

Hence

$$\begin{aligned}
\mathbf{Y}^T \mathbf{Y} &= \sigma^2 \mathbf{Z}^T \left(\boldsymbol{\Sigma}^{\frac{1}{2}}\right)^T \boldsymbol{\Sigma}^{\frac{1}{2}} \mathbf{Z} \\
&= \sigma^2 \mathbf{Z} \mathbf{A}^T \mathbf{A} \mathbf{Z} \\
&= \sigma^2 \mathbf{Z}^T \mathbf{A} \mathbf{Z}
\end{aligned}$$

Hence it suffices to show that

$$\mathbf{Z}^T \mathbf{A} \mathbf{Z} \sim \chi_{n-1}^2$$

Note that if $\mathbf{A} = \mathbf{I}$, then $\mathbf{Z}^T \mathbf{Z} \sim \chi_n^2$.

By spectral decomposition,

$$\mathbf{Z} \mathbf{A}^T \mathbf{Z} = \mathbf{Z} \mathbf{P} \boldsymbol{\Lambda} \mathbf{P}^T \mathbf{Z}$$

Denote $\mathbf{P}^T \mathbf{Z} = \mathbf{W} \in \mathbb{R}^n$.

$$\mathbf{Z} \mathbf{A}^T \mathbf{Z} = \mathbf{Z} \mathbf{P} \boldsymbol{\Lambda} \mathbf{P}^T \mathbf{Z} = \mathbf{W}^T \boldsymbol{\Lambda} \mathbf{W}$$

Because of spectral decomposition, we know that $\mathbf{P}$ and $\mathbf{P}^T$ are orthogonal matrices whose product is the identity. Applying an orthogonal matrix to a multivariate standard normal, i.g. $\mathbf{A}\mathbf{Z}$, does not change the multivariate standard normal distribution.

$$\mathbf{W} \sim N_n(\mathbf{0}, \mathbf{P} \mathbf{P}^T) = N_n(\mathbf{0}, \mathbf{I})$$

Hence

$$\begin{aligned}
\mathbf{W}^T \boldsymbol{\Lambda} \mathbf{W} &= \mathbf{W}^T \begin{pmatrix} 1 & 0 & \dots & 0 & 0 \\ 0 & 1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & 0 \\ 0 & 0 & \dots & 1 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{pmatrix} \mathbf{W} \\
&= \sum_{i=1}^{n-1} W_i^2 \\
&\sim \chi_{n-1}^2 \quad \checkmark
\end{aligned}$$

✓

We know the marginal distributions for the sample mean and variance

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right), s^2 \sim \frac{\sigma^2}{n-1} \chi_{n-1}^2$$

Result 5.5. If $X_1, X_2, \dots, X_n$ iid normal, $\bar{X}, s^2$ independent.

Proof. Define

$$\mathbf{Z} = \begin{pmatrix} X_1 - \bar{X} \\ X_2 - \bar{X} \\ \vdots \\ X_n - \bar{X} \\ \bar{X} \end{pmatrix}$$

$$\boldsymbol{Z} = \boldsymbol{B}\boldsymbol{X}, \boldsymbol{B} \in \mathbb{R}^{(n+1) \times n}, \boldsymbol{B} = \begin{pmatrix} & & & \\ & & & \\ & & A & \\ & & & \\ \frac{1}{2} & -\frac{1}{2} & \cdots & \frac{1}{2} \end{pmatrix}$$

Hence $\mathbf{Z}$ is $(n + 1)$ -dimensional normal.

If we show that $X_1 - \bar{X} \perp \bar{X}$, $X_2 - \bar{X} \perp \bar{X} \dots X_n - \bar{X} \perp \bar{X}$, then

$$\begin{pmatrix} X_1 - \bar{X} \\ X_2 - \bar{X} \\ \vdots \\ X_n - \bar{X} \end{pmatrix} \perp \bar{X} \implies s^2 \perp \bar{X}$$

Since $\mathbf{Z} = \mathbf{B}\mathbf{X}$ is multivariate normal, it suffices to check that the covariance is 0.

To show $Cov(\bar{X}, X_i - \bar{X}) = 0$ for all $1 \leq i \leq n$.

Note that

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \sum_{i=1}^n a_i X_i, \quad a_1 = a_2 = \dots a_n = \frac{1}{n}$$

$$X_1 - \bar{X} = \sum_{i=1}^n b_i X_i, \quad b_1 = 1 - \frac{1}{n}, b_2 = b_3 = \dots b_n = -\frac{1}{n}$$

$$\begin{aligned} Cov\left(\sum_{i=1}^n a_i X_i, \sum_{i=1}^n b_i X_i\right) &= \sum_{i=1}^n \sum_{j=1}^n a_i b_j Cov(X_i X_j) \\ &= \sum_{i=1}^n a_i b_i Var(X_i) \\ &= \sigma^2 \sum_{i=1}^n a_i b_i \\ &= 0 \end{aligned}$$

Remark. As a general strategy, to show independence, we can show in two steps

1. show jointly normal
2. show 0 covariance

6 Class 6

6.1 Basic Framework of Statistical Estimation

Given iid samples $X_1, X_2, \dots, X_n$, how do we infer / estimate parameters of F ?

Example. (Bernoulli): Given $X_1, X_2, \dots, X_n$ iid $Ber(p)$,

$$X_i = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

An estimate of p is

$$\hat{p} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{T}{n}$$

Where T is the number of heads and $T \sim Binom(n, p)$.

Example. (Normal): $X_1, X_2, \dots, X_n$ iid $N(\mu, \sigma^2)$. Estimates for parameters are

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

6.2 Properties of Estimators

Definition 6.1. (Unbiasedness): An estimate $\hat{\theta}$ is said to be unbiased for a parameter θ if

$$\mathbb{E}[\hat{\theta}] = \theta$$

for all values of the population.

Example. (Bernoulli):

$$\mathbb{E}[\hat{p}] = \frac{\mathbb{E}[T]}{n} = \frac{np}{n} = p$$

Hence $\hat{p}$ is an unbiased estimate of p .

Note that $\hat{p}^2$ is not an unbiased estimate of p^2 .

$$\begin{aligned} \mathbb{E}[\hat{p}^2] &= \mathbb{E}[T^2] \left(\frac{1}{n^2} \right) \\ &= (np(1-p) + n^2p^2) \left(\frac{1}{n^2} \right) \\ &= p^2 + \frac{p(1-p)}{n} \\ &\neq p^2 \end{aligned}$$

Note that we can rearrange terms to get

$$\begin{aligned} \mathbb{E}[T^2] &= n^2p^2 + np(1-p) \\ &= (n^2 - n)p^2 + np \\ \frac{\mathbb{E}[T^2]}{n(n-1)} &= p^2 + \frac{p}{n-1} \end{aligned}$$

We try estimating $\frac{p}{n-1}$ by $\frac{T}{(n-1)n}$.

Consider the estimate

$$\tilde{p} = \frac{T^2}{n(n-1)} - \frac{T}{n(n-1)} = \frac{T(T-1)}{n(n-1)}$$

The expectation is

$$\mathbb{E}[\tilde{p}] = p^2 + \frac{p}{n-1} - \frac{p}{n-1} = p^2$$

Note that (Proof left as exercise)

$$\mathbb{E}\left[\frac{T(T-1)}{n(n-1)}\right] = \sum_{r=2}^n \frac{r(r-1)}{n(n-1)} \binom{n}{r} p^r (1-p)^{n-r} = p^2$$

In general, an unbiased estimate for p^k is

$$\frac{T(T-1)(T-2)\dots(T-k+1)}{n(n-1)(n-2)\dots(n-k+1)}$$

An unbiased estimate of $2p^2 + 5p^3$ is

$$2\frac{T(T-1)}{n(n-1)} + 5\frac{T(T-1)(T-2)}{n(n-1)(n-2)}$$

Example. (Normal): Estimate σ^2 with

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi_{n-1}^2 \left(\frac{\sigma^2}{n-1} \right)$$

The expectation of sample variance is

$$\begin{aligned} \mathbb{E}[s^2] &= \frac{\sigma^2}{n-1} \mathbb{E}[\chi_{n-1}^2] \\ &= \frac{\sigma^2}{n-1} (n-1) \\ &= \sigma^2 \end{aligned}$$

Example. (Network sampling)

A *network* refers to a graph of vertices that are connected if they have an interaction.

Network sampling helps with understanding *how networks look* by studying a small section of the network, and understanding *features of a large unobserved network* from a sample subgraph.

Motifs refer to patterns of small subgraphs, such as an edge, or a triangle.

Motif estimation refers to estimating the number of a particular type of motif, based on an observed sample subgraph.

Let G_n be a population graph on n -vertices. Our subgraph sampling model involves sampling each vertex of G_n with probability $p \in (0, 1)$ independently, and then observing the subgraph on the set of sampled vertices.

Goal: estimate the number of edges in G_n based on the observed graph.

- initial guess: count the number of edges in the observed graph
- Denote $\hat{E}(G_n)$ as the number of edges in observed graph
- Denote $E(G_n)$ as number of edges in population graph

Result:

$$\mathbb{E}[\hat{E}(G_n)] = p^2$$

Hence,

$$\frac{\hat{E}(G_n)}{p^2} = \frac{\# \text{ edges in observed graph}}{p^2}$$

is an unbiased estimate of the number of edges in the population.

proof: Note that

$$E(G_n) = \sum_{1 \leq i \leq j \leq n} a_{ij}$$

Where

$$a_{ij} = \begin{cases} 1 & \text{if } (i, j) \text{ edge in } G_n \\ 0 & \text{otherwise} \end{cases}$$

Denote S the set of sampled vertices.

$$\begin{aligned}
\hat{E}(G_n) &= \sum_{1 \leq i \leq j \leq n, i \in S, j \in S} a_{ij} \\
&= \sum_{1 \leq i \leq j \leq n} a_{ij} \mathbf{1}[i \in S] \mathbf{1}[j \in S] \\
\mathbb{E} [\hat{E}(G_n)] &= a_{ij} \sum_{1 \leq i \leq j \leq n} \mathbb{E} [\mathbf{1}[i \in S] \mathbf{1}[j \in S]] \\
&= \sum_{1 \leq i \leq j \leq n} a_{ij} \mathbb{P}(i \in S) \mathbb{P}(j \in S) \\
&= p^2 \sum_{1 \leq i \leq j \leq n} a_{ij} \\
&= p^2 E(G_n)
\end{aligned}$$

Definition 6.2. (Variance of an estimate): The variance of an estimate $\hat{\theta}$ is

$$Var(\hat{\theta}) = \mathbb{E} \left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}] \right)^2 \right]$$

Example. (Bernoulli):

$$Var(\hat{p}) = Var \left(\frac{Binom(n, p)}{n} \right) = \frac{np(1-p)}{n^2} = \frac{p(1-p)}{n}$$

Example. (Normal):

$$\begin{aligned}
Var(\hat{\mu}) &= Var(\bar{X}) = \frac{\sigma^2}{n} \\
Var(\hat{\sigma}^2) &= Var(s^2) \\
&= Var \left(\frac{\sigma^2}{n-1} \chi_{n-1}^2 \right) \\
&= \left(\frac{\sigma^2}{n-1} \right)^2 Var(\chi_{n-1}^2) \\
&= \frac{\sigma^4}{(n-1)^2} 2(n-1) \\
&= \frac{2\sigma^4}{n-1}
\end{aligned}$$

Definition 6.3. Definition (Mean Squared Error): The mean squared error of an estimate $\hat{\theta}$ is

$$MSE(\hat{\theta}) = \mathbb{E} \left[\left(\hat{\theta} - \theta \right)^2 \right]$$

Result 6.4.

$$MSE(\hat{\theta}) = (Bias(\hat{\theta}))^2 + Var(\hat{\theta})$$

Proof.

Proof:

$$\begin{aligned}MSE(\hat{\theta}) &= \mathbb{E} \left[\left(\hat{\theta} - \theta \right)^2 \right] \\&= \mathbb{E} \left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}] + \mathbb{E}[\hat{\theta}] - \theta \right)^2 \right] \\&= \mathbb{E} \left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}] \right)^2 + 2 \left(\hat{\theta} - \mathbb{E}[\hat{\theta}] \right) \left(\mathbb{E}[\hat{\theta}] - \theta \right) + \left(\mathbb{E}[\hat{\theta}] - \theta \right)^2 \right] \\&= \mathbb{E} \left[\left(\hat{\theta} - \mathbb{E}[\hat{\theta}] \right)^2 \right] + 2 \left(\mathbb{E}[\hat{\theta}] - \theta \right) \mathbb{E} \left[\hat{\theta} - \mathbb{E}[\hat{\theta}] \right] + \left(\mathbb{E}[\hat{\theta}] - \theta \right)^2 \\&= \text{Var}(\hat{\theta}) + 0 + \left(\mathbb{E}[\hat{\theta}] - \theta \right)^2 \\&= \text{Var}(\hat{\theta}) + \left(\text{Bias}(\hat{\theta}, \theta) \right)^2\end{aligned}$$

✓

Corollary 6.5. If $\hat{\theta}$ unbiased, then $MSE(\hat{\theta}) = \text{Var}(\hat{\theta})$

7 Class 7

7.1 Consistency

Definition 7.1. (Consistency): Suppose $\hat{\theta}_n$ is an estimate of θ based on n iid samples. Then, $\hat{\theta}_n$ is said to be consistent for θ if

$$\hat{\theta}_n \xrightarrow{p} \theta$$

as $n \rightarrow \infty$

Result 7.2. MSE converging to 0 in the limit implies consistency

$$MSE(\hat{\theta}_n) \rightarrow 0 \implies \hat{\theta}_n \xrightarrow{p} \theta$$

Proof. By Markov's Inequality, for any $\epsilon > 0$

$$P\left(\left|\hat{\theta}_n - \theta\right| > \epsilon\right) \leq \frac{\mathbb{E}\left[\left(\hat{\theta}_n - \theta\right)^2\right]}{\epsilon^2}$$

✓

Remark. To prove consistency, we can

- Show MSE vanishes
- Use Continuous Mapping
- Use Slutsky's Lemma

Example. (Bernoulli):

$$Bias(\hat{p}) = 0, Var(\hat{p}) = \frac{p(1-p)}{n}$$

As $n \rightarrow \infty$

$$MSE(\hat{p}) = Var(\hat{p}) = \frac{p(1-p)}{n} \rightarrow 0$$

Example. (Normal):

$$\hat{\mu} = \bar{X} \implies bias(\hat{\mu}) = 0, Var(\hat{\mu}) = Var(\bar{X}) = \frac{\sigma^2}{n}$$

As $n \rightarrow \infty$

$$MSE(\bar{X}) = \frac{\sigma^2}{n} \rightarrow 0$$

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

$$bias(\hat{\sigma}^2) = 0, Var(\hat{\sigma}^2) = Var\left(\sigma^2 \frac{\chi_{n-1}^2}{n-1}\right) = \frac{2\sigma^4}{n-1}$$

As $n \rightarrow \infty$

$$MSE(\hat{\sigma}^2) = Var(\hat{\sigma}^2) \rightarrow 0$$

7.2 Method of moments estimation

Definition 7.3. (Method of Moments): Suppose $X_1, X_2, \dots, X_n$ iid from distribution F which has k unknown parameters $(\theta_1, \theta_2, \dots, \theta_k)$.

Denote $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_k)$.

For each $1 \leq j \leq k$, compute the j -th moment of F

$$\alpha_j(\boldsymbol{\theta}) = \mathbb{E}[X^j] = \int X_j f(x) dx$$

Define the j -th sample moment as

$$\hat{\alpha}_j = \frac{1}{n} \sum_{i=1}^n X_i^j$$

Set up k equations

$$\begin{cases} \alpha_1 = \hat{\alpha}_1 \\ \alpha_2 = \hat{\alpha}_2 \\ \vdots \\ \alpha_k = \hat{\alpha}_k \end{cases}$$

The method of moments estimate $\hat{\theta}_n$ is the solution to the system of equations.

Remark. Note that

- The above system is a system of k equations in k unknowns
- The above system may have 0, 1 or multiple solutions. If solution is unique, then the solution is the method of moments estimate of θ

Example. (Bernoulli): $X_1, \dots, X_n \sim \text{Ber}(p)$

$$\alpha_1 = \alpha_1(p) = \mathbb{E}[X] = p$$

$$\hat{\alpha}_1 = \frac{1}{n} \sum_{i=1}^n X_i$$

Solving for p :

$$\begin{aligned} \hat{\alpha}_1 &= \alpha_1 \\ \implies \hat{p} &= \frac{1}{n} \sum_{i=1}^n X_i \end{aligned}$$

Example. (Normal) $X_1, \dots, X_n \sim N(\mu, \sigma^2)$

The population moments are

$$\alpha_1 = \mu$$

$$\alpha_2 = \mathbb{E}[X^2] = \sigma^2 + \mu^2$$

Equating to sample moments

$$\mu = \alpha_1 = \hat{\alpha}_1 = \frac{1}{n} \sum_{i=1}^n X_i$$

$$\mu^2 + \sigma^2 = \alpha_2 = \hat{\alpha}_2 = \frac{1}{n} \sum_{i=1}^n X_i^2$$

Therefore,

$$\begin{aligned} \hat{\mu} &= \sum_{i=1}^n X_i \\ \hat{\sigma}^2 &= \frac{1}{n} \sum_{i=1}^n X_i^2 - \left(\frac{1}{n} \sum_{i=1}^n X_i \right)^2 \\ &= \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 \\ &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \end{aligned}$$

The last equality follows because

$$\begin{aligned}
& \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \\
&= \frac{1}{n} \sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2) \\
&= \frac{1}{n} \sum_{i=1}^n X_i^2 - \frac{2}{n} \bar{X} \sum_{i=1}^n X_i + \bar{X}^2 \\
&= \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2
\end{aligned}$$

Note: $\hat{\sigma}_{MLE}^2$ is a biased estimator of σ^2 .

$$\mathbb{E}[\hat{\sigma}^2] = \frac{n-1}{n} \mathbb{E}[s^2] = \frac{n-1}{n} \sigma^2$$

As $n \rightarrow \infty$

$$\begin{aligned}
Bias(\hat{\theta}^2) &= \frac{n-1}{n} \sigma^2 - \sigma^2 \rightarrow 0 \\
Var(\hat{\theta}^2) &= Var\left(\frac{n-1}{n} s^2\right) \\
&= \left(\frac{n-1}{n}\right)^2 \left(\frac{2\sigma^4}{n-1}\right) \\
&= \frac{n-1}{n^2} 2\sigma^4 \\
&\rightarrow 0
\end{aligned}$$

Hence $MSE(\hat{\sigma}) \rightarrow 0$, and $\hat{\sigma}_{MLE}^2$ is consistent despite being biased.

Example. Ex (Pearson, 1984).

Contributions to the mathematical theory of evolution (1894). Weldon (1893) crab data: body length. The histogram of the data did not look like the bell-shaped Gaussian curve.

Conjecture: maybe two species, each Gaussian? (Gaussian Mixture Model.)

Pearson conjectured that the data consists of two different kinds of crabs, each with its own Gaussian distribution.

Gaussian Mixture Model (GMM)

Let W be a random variable such that

$$W \sim \begin{cases} N(\mu_1, \sigma_1^2), & \text{with prob. } p, \\ N(\mu_2, \sigma_2^2), & \text{with prob. } (1-p). \end{cases}$$

Distribution Function

$$P(W \leq t) = P(W \leq t, Z = 1) + P(W \leq t, Z = 2),$$

where $Z = \begin{cases} 1 & \text{if group 1 is sampled,} \\ 2 & \text{if group 2 is sampled.} \end{cases}$

So,

$$P(W \leq t) = p P(N(\mu_1, \sigma_1^2) \leq t) + (1-p) P(N(\mu_2, \sigma_2^2) \leq t).$$

Note: GMM is *not* adding two normals. Adding two normals gets another normal, but GMM is not a sum. It is a mixture.

Density Function

$$f(t) = \frac{d}{dt} P(W \leq t) = p \phi\left(\frac{t - \mu_1}{\sigma_1}\right) + (1-p) \phi\left(\frac{t - \mu_2}{\sigma_2}\right),$$

where $\phi(t) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{t^2}{2}\right)$.

Estimation of Parameters of GMM

1. Compute sample moments:

$$\hat{m}_r = \frac{1}{n} \sum_{i=1}^n X_i^r, \quad 1 \leq r \leq 5.$$

2. Compute population moments:

$$m_r = \mathbb{E}[X^r], \quad 1 \leq r \leq 5.$$

3. Solve for $\mu_1, \mu_2, \sigma_1^2, \sigma_2^2, p$ from

$$\hat{m}_r = m_r, \quad r = 1, \dots, 5.$$

Pearson's Sixth-Moment Test

In general, the system of equations can have multiple roots. To choose a root, Pearson's approach: choose the root $(\hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}_1^2, \hat{\sigma}_2^2, \hat{p})$ that is closest to the sixth sample moment.

That is, the feasible root with the smallest value of

$$|\hat{m}_6 - m_6| = \left| \frac{1}{n} \sum_{i=1}^n X_i^6 - \mathbb{E}[X^6] \right|.$$

Theorem (Kalai, Moitra, Valiant, 2010): the sixth-moment method gives consistent estimates of the parameters of GMM and can be computed efficiently.

8 Class 8

8.1 Maximum Likelihood Estimation

Definition 8.1. (Maximum Likelihood Estimation): Let $X_1, \dots, X_n$ iid from F with *pdf* or *pmf* f_θ . Consider the joint *pdf* or *pmf*

$$L(\theta|x_1, x_2, \dots, x_n) = \prod_{i=1}^n f_\theta(x_i)$$

The maximum likelihood estimator is given by

$$\hat{\theta}_n = \arg \max_{\Theta} L(\theta|x_1, x_2, \dots, x_n)$$

Equivalently, define

$$\begin{aligned} l(\theta|x_1, \dots, x_n) &:= \log L(\theta|x_1, \dots, x_n) \\ &= \sum_{i=1}^n \log f_\theta(x_i) \\ \hat{\theta}_n &= \arg \max_{\Theta} l(\theta|x_1, x_2, \dots, x_n) \end{aligned}$$

Example. (Bernoulli) $X_1, \dots, X_n \sim \text{Ber}(p)$.

$$\begin{aligned} f_p(x) &= P(X = x) \\ &= \begin{cases} p & \text{if } x = 1 \\ 1 - p & \text{if } x = 0 \end{cases} \\ &= p^x(1 - p)^{1-x} \end{aligned}$$

The likelihood function is

$$\begin{aligned} L(p|X_1, \dots, X_n) &= \prod_{i=1}^n f_p(X_i) \\ &= \prod_{i=1}^n p^{X_i} (1 - p)^{1-X_i} \\ &= p^{\sum_{i=1}^n X_i} (1 - p)^{n - \sum_{i=1}^n X_i} \\ l(p|X_1, X_2, \dots, X_n) &= \left(\sum_{i=1}^n X_i \right) \log p + \left(n - \sum_{i=1}^n X_i \right) \log(1 - p) \end{aligned}$$

Define $T = \sum_{i=1}^n X_i$,

$$\begin{aligned} l(p|X_1, \dots, X_n) &= T \log p + (n - T) \log(1 - p) \\ \frac{d}{dp} l(p|X_1, \dots, X_n) &= \frac{T}{p} - \frac{n - T}{1 - p} \\ &= 0 \end{aligned}$$

Hence

$$\hat{p}_{MLE} = \frac{T}{n} = \frac{1}{n} \sum_{i=1}^n X_i$$

To show that this is indeed a maximum,

$$\frac{d^2}{dp^2} l(p|X_1, \dots, X_n) = -\frac{T}{p^2} - \frac{T}{(1 - p)^2} < 0$$

Hence $l(p|X_1, \dots, X_n)$ is concave and $\hat{p} = \frac{T}{n}$ is the unique maximum.

By LLN,

$$\hat{p} \xrightarrow{P} p$$

By CLT,

$$\sqrt{n}(\hat{p} - p) \xrightarrow{d} N(0, p(1-p))$$

Example. (Normal): $X_1, \dots, X_n$ iid $N(\mu, \sigma^2)$

$$\begin{aligned} L(\mu, \sigma^2 | X_1, \dots, X_n) &= \prod_{i=1}^n f_{\mu, \sigma^2}(X_i) \\ &= \prod_{i=1}^n \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) \exp \left(-\frac{1}{2\sigma^2} (X_i - \mu)^2 \right) \\ &= \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right)^n \exp \left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \right) \end{aligned}$$

The log likelihood is

$$\begin{aligned} l(\mu, \sigma^2 | X_1, \dots, X_n) &= \log(L(\mu, \sigma^2 | X_1, \dots, X_n)) \\ &= n \log \frac{1}{\sqrt{2\pi\sigma^2}} - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \\ &= -\frac{n}{2} \log 2\pi\sigma^2 - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \end{aligned}$$

Define

$$g(\mu, \sigma^2) := -\frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2$$

Setting first order partials to zero, we get

$$\begin{aligned} \frac{\partial g(\mu, \sigma^2)}{\partial \mu} &= \frac{1}{\sigma^2} \sum_{i=1}^n (X_i - \mu) = 0 \\ \frac{\partial g(\mu, \sigma^2)}{\partial \sigma^2} &= -\frac{n}{2} \left(\frac{1}{\sigma^2} \right) + \frac{1}{2\sigma^4} \sum_{i=1}^n (X_i - \mu)^2 = 0 \end{aligned}$$

Hence

$$\begin{aligned} \hat{\mu}_{MLE} &= \frac{1}{n} \sum_{i=1}^n X_i = \bar{X} \\ \hat{\sigma}_{MLE}^2 &= \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \end{aligned}$$

To show that this is the global max, we can either

- Find the Hessian and show that it is negative definite, i.e. the function is concave
- Argue that the derivative is 0 at only one point, and show that for extreme values of μ and σ^2 (i.e. $\mu \rightarrow -\infty$, $\mu \rightarrow \infty$, $\sigma^2 \rightarrow 0$, $\sigma^2 \rightarrow \infty$)

$$l(\mu, \sigma^2) \rightarrow -\infty$$

Note

1. $\hat{\mu} \rightarrow \mu$ by LLN
2. $\sqrt{n}(\hat{\mu} - \mu) \xrightarrow{d} N(0, \sigma^2)$
3. $\hat{\sigma}^2 \xrightarrow{P} \sigma^2$ by computing MSE.
4. $\sqrt{n}(\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} N(0, 2\sigma^4)$

Proof of 4:

$$\begin{aligned} \sqrt{n}(\hat{\sigma}^2 - \sigma^2) &= \sqrt{n} \left(\frac{\sigma^2}{n-1} \chi_{n-1}^2 - \sigma^2 \right) \\ &= \sigma^2 \sqrt{n} \left(\frac{\chi_{n-1}^2}{n-1} - 1 \right) \end{aligned}$$

Note that

$$\sqrt{n} \left(\frac{\chi_n^2}{n} - 1 \right) \xrightarrow{d} N(0, 2)$$

Hence

$$\sqrt{n} (\hat{\sigma}^2 - \sigma^2) \xrightarrow{d} N(0, 2\sigma^4)$$

REVISIT: check with the professor regarding this proof on asymptotic normality of $\hat{\sigma}_{MLE}^2$ without using Fisher Information.

9 Class 9

9.1 Fisher Information

Definition 9.1. (Score Function): The score function is defined as

$$s(\theta) = \frac{d}{d\theta} \log f_{\theta}(X)$$

Definition 9.2. Definition (Fisher Information): The Fisher Information is defined as

$$I(\theta) = \mathbb{E}[(s(\theta))^2] = \mathbb{E}\left[\left(\frac{d}{d\theta} \log f_{\theta}(X)\right)^2\right]$$

Lemma 9.3. Lemma (Properties of the score function):

1. Zero expectation

$$\mathbb{E}[s(\theta)] = \mathbb{E}\left[\frac{d}{d\theta} \log f_{\theta}(X)\right] = 0$$

2. The variance is Fisher Information

$$\text{Var}\left(\frac{d}{d\theta} \log f_{\theta}(x)\right) = I(\theta)$$

3. The expectation of the second derivative is negative Fisher information

$$I(\theta) = -\mathbb{E}\left[\frac{d^2}{d\theta^2} \log f_{\theta}(X)\right]$$

Proof. (1):

$$\begin{aligned}\mathbb{E}\left[\frac{d}{d\theta} \log f_{\theta}(X)\right] &= \int \frac{d}{d\theta} \log f_{\theta}(x) f_{\theta}(x) dx \\ &= \int \frac{\frac{d}{d\theta} f_{\theta}(x)}{f_{\theta}(x)} f_{\theta}(x) dx \\ &= \int \frac{d}{d\theta} f_{\theta}(x) dx \\ &= \frac{d}{d\theta} \int f_{\theta}(x) dx \\ &= \frac{d}{d\theta} 1 \\ &= 0\end{aligned}$$

(2): Proof is immediate

$$\begin{aligned}\text{Var}(s(\theta)) &= \mathbb{E}[(s(\theta))^2] - \mathbb{E}[s(\theta)]^2 \\ &= \mathbb{E}[(s(\theta))^2] \\ &= I(\theta)\end{aligned}$$

✓

9.2 Asymptotic properties of the MLE estimator

Result 9.4. (Asymptotic properties of MLE): Suppose $X_1, \dots, X_n$ iid with pdf or pmf $f_{\theta_0}(x)$, $\theta_0 \in \Omega$ where θ_0 is the true parameter and Ω is the parameter space.

Denote by $\hat{\theta}$ the MLE estimate based on $X_1, \dots, X_n$.

Suppose the following assumptions

- The density / pmf f_{θ} has the same support for all $\theta \in \Omega$, i.e.

$$\{x : f_{\theta}(x) > 0\}$$

does not depend on θ

- θ_0 is an interior point of Ω
- The log-likelihood function $l_n(\theta) = l(\theta|X_1, \dots, X_n)$ is differentiable in θ
- $\hat{\theta}$ is the unique value of $\theta \in \Omega$ such that

$$l'_n(\theta) = 0$$

Then the following hold

1. $\hat{\theta}$ consistent for θ_0 , i.e. as $n \rightarrow \infty$

$$\hat{\theta} \rightarrow \theta_0$$

- 2.

and

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, \frac{1}{I(\theta_0)}\right)$$

Proof. Normality implies consistency: if we know that the asymptotic distribution is normal with variance being the inverse of Fisher Information, then

$$(\hat{\theta} - \theta_0) = \frac{\sqrt{n}(\hat{\theta} - \theta_0)}{\sqrt{n}} \xrightarrow{p} 0 \text{ by Slutsky's}$$

Asymptotic normality:

$$l'_n(\hat{\theta}) = 0$$

By Taylor Expansion around θ_0 ,

$$\begin{aligned} l'_n(\hat{\theta}) &\approx l'_n(\theta_0) + (\hat{\theta} - \theta_0) l''_n(\theta_0) \\ \implies (\hat{\theta} - \theta_0) &\approx \frac{-l'_n(\theta_0)}{l''_n(\theta_0)} \\ \implies \sqrt{n}(\hat{\theta} - \theta_0) &\approx \frac{-\frac{1}{\sqrt{n}}l'_n(\theta_0)}{\frac{1}{n}l''_n(\theta_0)} \end{aligned}$$

Consider the denominator

$$\begin{aligned} &\frac{1}{n}l''_n(\theta_0) \\ &= \frac{1}{n} \sum_{i=1}^n \frac{\partial^2}{\partial \theta^2} \log f_\theta(X_i) \Big|_{\theta=\theta_0} \\ &\xrightarrow{p} \mathbb{E} \left[\frac{\partial^2}{\partial \theta^2} \log f_\theta(X_i) \Big|_{\theta=\theta_0} \right] \text{ by law of large numbers} \\ &= -I(\theta_0) \end{aligned}$$

Consider the numerator

$$\begin{aligned} &\frac{1}{\sqrt{n}}l'_n(\theta_0) \\ &= \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f_\theta(X_i) \Big|_{\theta=\theta_0} \\ &= \sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n \frac{\partial}{\partial \theta} \log f_\theta(X_i) \Big|_{\theta=\theta_0} \right) \\ &\xrightarrow{d} N(0, I(\theta_0)) \text{ by CLT} \end{aligned}$$

Hence

$$\begin{aligned}
& \sqrt{n} (\hat{\theta} - \theta_0) \\
& \approx -\sqrt{n} \left(\frac{l'_n(\theta_0)}{l''_n(\theta_0)} \right) \\
& \approx \frac{-\frac{1}{\sqrt{n}} l'_n(\theta_0)}{\frac{1}{n} l''_n(\theta_0)} \\
& \xrightarrow{d} \frac{N(0, I(\theta_0))}{I(\theta_0)} \text{ by Slutsky's} \\
& = N\left(0, \frac{1}{I(\theta_0)}\right)
\end{aligned}$$

✓

Example. (Bernoulli): $X_1, x_2, \dots, X_n \sim \text{Ber}(P)$. Then,

$$\begin{aligned}
f_p(x) &= p^x (1-p)^{1-x} \\
\log f_p(x) &= x \log p + (1-x) \log(1-p) \\
\frac{\partial}{\partial p} \log f_p(x) &= \frac{x}{p} - \frac{1-x}{1-p} \\
-\frac{\partial^2}{\partial p^2} \log f_p(x) &= \frac{x}{p^2} + \frac{1-x}{(1-p)^2}
\end{aligned}$$

Therefore

$$\begin{aligned}
I(p) &= -\mathbb{E} \left[\left(\frac{\partial}{\partial p} \log f_p(x) \right)^2 \right] \\
&= \frac{p}{p^2} + \frac{1-p}{(1-p)^2} \\
&= \frac{1}{p} + \frac{1}{1-p} \\
&= \frac{1}{p(1-p)}
\end{aligned}$$

Hence

$$\begin{aligned}
\sqrt{n} (\hat{p}_{MLE} - p) &\xrightarrow{p} N\left(0, \frac{1}{I(p)}\right) \\
&= N(0, p(1-p))
\end{aligned}$$

Example. (Normal): $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2)$. The pdf of X is

$$\begin{aligned}
f_{\mu, \sigma^2}(x) &= \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2} \\
\log f_{\mu, \sigma^2}(x) &= -\frac{1}{2} \log 2\pi - \frac{1}{2} \log \sigma^2 - \frac{1}{2\sigma^2}(x-\mu)^2
\end{aligned}$$

Fisher information for μ :

$$\begin{aligned}
\frac{\partial}{\partial \mu} \log f(x) &= \frac{1}{\sigma^2}(x-\mu) \\
-\frac{\partial^2}{\partial \mu^2} \log f(x) &= \frac{1}{\sigma^2}
\end{aligned}$$

Therefore,

$$I(\mu) = \frac{1}{\sigma^2}$$

Hence

$$\sqrt{n}(\hat{\mu}_{MLE} - \mu) \xrightarrow{d} N(0, \sigma^2)$$

Fisher information for σ^2 :

$$\begin{aligned}
\frac{\partial}{\partial \sigma^2} \log f(x) &= -\frac{1}{2\sigma^2} + \frac{1}{2\sigma^4}(x-\mu)^2 \\
-\frac{\partial^2}{(\partial \sigma^2)^2} \log f(x) &= -\frac{1}{2\sigma^4} + \frac{1}{\sigma^6}(x-\mu)^2
\end{aligned}$$

Therefore,

$$I(\sigma^2) = -\mathbb{E} \left[\frac{\partial^2}{(\partial \sigma^2)^2} \right] = -\frac{1}{2\sigma^4} + \frac{1}{\sigma^4} = \frac{1}{2\sigma^4}$$

And

$$\begin{aligned} \sqrt{n}(\hat{\sigma}_{MLE} - \sigma^2) &\xrightarrow{d} N\left(0, \frac{1}{I(\sigma^2)}\right) \\ &= N(0, 2\sigma^4) \end{aligned}$$

Remark. As a heuristic

$$\hat{\theta} \approx N\left(\theta_0, \frac{1}{nI(\theta)}\right)$$

9.3 Asymptotic properties of the MLE estimator for multiple parameters

Let $\{f_{\theta}(x) : \theta \in \Omega\}$ be a family of distributions where $\theta = (\theta_1, \theta_2, \dots, \theta_k) \in \mathbb{R}^k$ is a k -dimensional parameter.

Theorem 9.5. Suppose $X_1, \dots, X_n$ iid from distribution with density / pmf f_{θ} . Let $\hat{\theta}_n$ be the MLE based on $X_1, X_2, \dots, X_n$.

Let the $k \times k$ Fisher Information matrix be defined as

$$(I(\theta))_{ij} = Cov\left(\frac{\partial}{\partial \theta_i} \log f_{\theta}(x), \frac{\partial}{\partial \theta_j} \log f_{\theta}(x)\right)$$

Under the same conditions as the univariate normal,

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} N_k(\mathbf{0}, I(\theta)^{-1})$$

Remark. As an heuristic,

$$\hat{\theta}_n \approx N\left(\theta, \frac{1}{n} I(\theta)^{-1}\right)$$

Proof. We WTS that

$$Cov\left(\frac{\partial}{\partial \theta_i} \log f_{\theta}(x), \frac{\partial}{\partial \theta_j} \log f_{\theta}(x)\right) = -\mathbb{E} \left[\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f_{\theta}(x) \right]$$

✓

Example. (Normal):

$$\begin{aligned} -\frac{\partial^2}{\partial \mu \partial \sigma^2} l_n(\mu, \sigma^2) &= \frac{1}{\sigma^4} (x - \mu) \\ \mathbb{E} \left[-\frac{\partial^2}{\partial \mu \partial \sigma^2} l_n(\mu, \sigma^2) \right] &= 0 \end{aligned}$$

The Fisher Information matrix is

$$I(\mu, \sigma^2) = \begin{pmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{pmatrix}$$

Hence

$$I(\mu, \sigma^2)^{-1} = \begin{pmatrix} \sigma^2 & 0 \\ 0 & 2\sigma^4 \end{pmatrix}$$

and

$$\sqrt{n} \begin{pmatrix} \hat{\mu}_{MLE} - \mu \\ \hat{\sigma}_{MLE}^2 - \sigma^2 \end{pmatrix} = N_2 \left(\mathbf{0}, \begin{pmatrix} \sigma^2 & 0 \\ 0 & 2\sigma^4 \end{pmatrix} \right)$$

Example. (Gamma distribution)

Let $X_1, \dots, X_n$ iid $\Gamma(\alpha, \beta)$ with density

$$f_{\alpha, \beta}(x) = \frac{x^{\alpha-1} e^{-\frac{x}{\beta}}}{\beta^{\alpha} \Gamma(\alpha)}$$

The likelihood function is

$$\begin{aligned} L_n(\alpha, \beta) &= \prod_{i=1}^n f_{\alpha, \beta}(X_i) \\ &= \end{aligned}$$

The likelihood function is

$$L_n(\alpha, \beta) = \prod_{i=1}^n f_{\alpha, \beta}(X_i) = \frac{1}{(\beta^\alpha \Gamma(\alpha))^n} \prod_{i=1}^n X_i^{\alpha-1} e^{-\frac{1}{\beta} \sum_{i=1}^n X_i}.$$

The log-likelihood function is

$$\ell_n(\alpha, \beta) = \log L_n(\alpha, \beta) = -n\alpha \log \beta - n \log \Gamma(\alpha) + (\alpha - 1) \sum_{i=1}^n \log X_i - \frac{1}{\beta} \sum_{i=1}^n X_i.$$

Taking derivatives with respect to α and β and setting them to zero:

$$\frac{\partial \ell_n(\alpha, \beta)}{\partial \alpha} = -n \log \beta - n \frac{\Gamma'(\alpha)}{\Gamma(\alpha)} + \sum_{i=1}^n \log X_i = 0, \quad (1)$$

$$\frac{\partial \ell_n(\alpha, \beta)}{\partial \beta} = -\frac{n\alpha}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^n X_i = 0. \quad (2)$$

From (2), the MLE satisfies

$$\hat{\beta} = \frac{\bar{X}}{\hat{\alpha}}, \quad \text{where } \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i.$$

Substituting $\hat{\beta}$ into (1) and dividing by n , we obtain

$$\log \hat{\alpha} - \frac{\Gamma'(\hat{\alpha})}{\Gamma(\hat{\alpha})} - \log \bar{X} + \frac{1}{n} \sum_{i=1}^n \log X_i = 0.$$

Consider the function

$$f(\alpha) = \log \alpha - \frac{\Gamma'(\alpha)}{\Gamma(\alpha)}.$$

The function $f(\alpha)$ decreases from ∞ to 0 as α increases from 0 to ∞ .

Moreover,

$$\log \bar{X} - \frac{1}{n} \sum_{i=1}^n \log X_i > 0 \quad (\text{by Jensen's inequality}).$$

Hence,

$$\log \hat{\alpha} - \frac{\Gamma'(\hat{\alpha})}{\Gamma(\hat{\alpha})} = \log \bar{X} - \frac{1}{n} \sum_{i=1}^n \log X_i$$

has a unique solution $\hat{\alpha}$, which is the MLE of α .

10 Class 10

10.1 Cramer-Rao Lower Bound

Theorem 10.1. (Cramer-Rao Lower Bound) Consider a parametric model of distributions $\{f_\theta(x), \theta \in \Omega\}$ satisfying certain *mild regularity conditions*, and T is any unbiased estimator θ based on $X_1, X_2, \dots, X_n$ iid. Then

$$\text{Var}(T) \geq \frac{1}{nI(\theta)}$$

Proof. Denote the score function

$$s(\theta, x) = \frac{\partial}{\partial \theta} \log f_\theta(x) = \frac{\frac{\partial}{\partial \theta} f_\theta(x)}{f_\theta(x)}$$

Let

$$S = s(X_1, X_2, \dots, X_n) = \sum_{i=1}^n s(\theta, X_i)$$

For any unbiased estimator T ,

$$\begin{aligned} (\text{Cov}(S, T))^2 &\leq \text{Var}(S) \cdot \text{Var}(T) \\ \implies \text{Cov}(S, T)^2 &\leq \text{Var}(S) \cdot \text{Var}(T) \\ \implies \text{Var}(T) &\geq \frac{(\text{Cov}(S, T))^2}{\text{Var}(S)} \end{aligned}$$

Where

$$\begin{aligned} \text{Var}(S) &= n\text{Var}(s) \\ &= n\text{Var}\left(\frac{\partial}{\partial \theta} \log f_\theta(X_1)\right) \\ &= nI(\theta) \end{aligned}$$

Hence

$$\text{Var}(T) \geq \frac{(\text{Cov}(S, T))^2}{nI(\theta)}$$

To show that $\text{Cov}(S, T) = 1$, by unbiasedness of T

$$\begin{aligned} \theta &= \mathbb{E}[T] \\ &= \int_{\mathbb{R}^n} T(X_1, X_2, \dots, X_n) f_\theta(X_1) f_\theta(X_2) \dots f_\theta(X_n) dX_1 dX_2 \dots dX_n \end{aligned}$$

Taking the derivative wrt θ on both sides,

$$\begin{aligned} 1 &= \int_{\mathbb{R}^n} T(X_1, X_2, \dots, X_n) \left[\sum_{i=1}^n \left(\frac{\partial}{\partial \theta} f_\theta(x_i) \prod_{j \in [1, n], j \neq i} f_\theta(x_j) \right) \right] dX_1, dX_2 \dots dX_n \\ &= \int_{\mathbb{R}^n} T(X_1 \dots X_n) S(X_1, \dots, X_n) f_\theta(X_1) f_\theta(X_2) \dots f_\theta(X_n) dX_1 dX_2 \dots dX_n \\ &= \mathbb{E}_\theta[TS] \end{aligned}$$

Since the score function has zero expectation,

$$1 = \mathbb{E}[TS] - \mathbb{E}[T] \mathbb{E}[S] = \text{Cov}(S, T)$$

✓

Remark. An unbiased estimator is said to be efficient if its variance is

$$\frac{1}{nI(\theta)}$$

Remark. Since the MLE is asymptotically unbiased and the variance of the MLE attains the Cramer-Rao Lower Bound asymptotically, the MLE is said to be **asymptotically efficient**.

10.2 Simple Linear Regression

10.2.1 Estimating β_0, β_1

Definition 10.2. (Simple Linear Regression) Suppose we have data $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ related by model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

The **least squares estimates** are obtained via

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

The **fitted values** are defined as

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

The **residuals** are defined as

$$E_i = Y_i - \hat{Y}_i$$

The **sum of squared errors, SSE** are

$$SSE := \sum_{i=1}^n E_i^2 = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

Definition 10.3. Define

$$S_{xy} = \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = (n-1)s_{xy}$$

$$S_{xx} = \sum_{i=1}^n (x_i - \bar{x})^2 = (n-1)s_x^2$$

$$S_{yy} = \sum_{i=1}^n (y_i - \bar{y})^2 = (n-1)s_y^2$$

Remark. Define

$$f(\beta_0, \beta_1) = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2$$

The LS estimates $(\hat{\beta}_0, \hat{\beta}_1)$ are solved via

$$\begin{cases} \frac{\partial f}{\partial \beta_0} = 0 \\ \frac{\partial f}{\partial \beta_1} = 0 \end{cases}$$

Result 10.4. The LS line always passes through the point $(\bar{X}, \bar{Y})$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

Remark. (Derivation of regression parameters)

Let

$$f(a, b) = \sum_{i=1}^n (y_i - (a + bx_i))^2$$

Setting the partial derivatives to 0

$$\begin{aligned}\frac{\partial f}{\partial a} &= -2 \sum_{i=1}^n (y_i - (a + bx_i)) = 0 \\ \frac{\partial f}{\partial b} &= -2 \sum_{i=1}^n (y_i - (a + bx_i)) x_i = 0\end{aligned}$$

This is a system of linear equations in a, b .

$$\begin{aligned}(2n)a + \left(2 \sum_{i=1}^n x_i\right)b &= 2 \sum_{i=1}^n y_i \\ \left(2 \sum_{i=1}^n x_i\right)a + \left(2 \sum_{i=1}^n x_i^2\right)b &= 2 \sum_{i=1}^n x_i y_i\end{aligned}$$

Solving for a, b

$$\begin{aligned}\hat{\beta}_0 = a &= \frac{(\sum x_i^2)(\sum y_i) - (\sum x_i)(\sum x_i y_i)}{n \sum x_i^2 - (\sum x_i)^2} \\ \hat{\beta}_1 = b &= \frac{n \sum x_i y_i - (\sum x_i)(\sum y_i)}{n \sum x_i^2 - (\sum x_i)^2}\end{aligned}$$

Also

$$\begin{aligned}\hat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} = \frac{s_{xy}}{s_x^2} = r \frac{s_y}{s_x} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}\end{aligned}$$

Remark. Note that the results thus far assumes nothing about the ϵ_i 's.

10.2.2 Estimating variance

Now assume that

$$\epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2)$$

By definition

$$\begin{aligned}\epsilon_i &= Y_i - \beta_0 - \beta_1 X_i \\ \|\epsilon\|^2 &= \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2\end{aligned}$$

If β_0, β_1 known, then

$$\mathbb{E} \left[\frac{1}{n} \sum_{i=1}^n \epsilon_i^2 \right] = \sigma^2$$

and an unbiased estimator of σ^2 is

$$\frac{\|\epsilon\|^2}{n}$$

Result 10.5. $\hat{\sigma}^2$ defined below is an unbiased estimate of σ^2

$$SSE \sim \sigma^2 \chi_{n-2}^2 \implies \hat{\sigma}^2 = \frac{SSE}{n-2}$$

10.2.3 Facts about Least Square Estimates

Result 10.6. The LS estimates are unbiased.

$$\begin{aligned}\mathbb{E} [\hat{\beta}_1] &= \beta_1 \\ \mathbb{E} [\hat{\beta}_0] &= \beta_0\end{aligned}$$

Result 10.7. The LS estimates are normally distributed.

$$\hat{\beta}_0 \sim N \left(\beta_0 \frac{\sigma^2 \sum_{i=1}^n x_i^2}{n s_{xx}}, \frac{\sigma^2}{n s_{xx}} \right)$$

$$\hat{\beta}_1 \sim N \left(\beta_1, \frac{\sigma^2}{s_{xx}} \right)$$

where

$$s_{xx} = \sum_{i=1}^n (x_i - \bar{X})^2$$

Result 10.8. The LS estimates find the vector $\hat{\mathbf{v}}$ in the plane spanned by the vectors $\mathbf{1}, \mathbf{X}$ that is the closest to $\mathbf{Y}$, where

$$\hat{\mathbf{v}} = \hat{\beta}_0 \mathbf{1} + \hat{\beta}_1 \mathbf{X}$$

To see this, denote $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$ with $v_i = \beta_0 + \beta_1 x_i$. This can be written as

$$\mathbf{v} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$$

where

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Therefore, $\mathbf{v}$ is a linear combination of $\mathbf{1}$ and $\mathbf{x}$.

The LS estimate minimizes

$$\sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2 = \|\mathbf{Y} - \mathbf{v}\|^2$$

Result 10.9. $\hat{\beta}_0, \hat{\beta}_1$ are independent of $\hat{\sigma}^2$.

Hence

$$\frac{\hat{\beta}_1 - \beta_1}{S/\sqrt{s_{xx}}} \sim t_{n-2}$$

$$\frac{\hat{\beta}_0 - \beta_0}{S\sqrt{\frac{\sum x_i^2}{n s_{xx}}}} \sim t_{n-2}$$

This can be used for constructing confidence intervals and hypothesis testing for β_0 and β_1 .

Result 10.10. $\hat{\beta}_0, \hat{\beta}_1$ coincides with the MLE estimates.

Note that $Y_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$. The likelihood function is

$$\left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2 \right\}$$

The log likelihood function is

$$-\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 x_i)^2 - \frac{n}{2} \log(2\pi\sigma^2)$$

The MLE is obtained by

$$\arg \min_{\beta_0, \beta_1, \sigma^2} l(\beta_0, \beta_1, \sigma^2)$$

11 Class 11

11.1 Least Absolute Deviation Line

Definition 11.1. (Least Absolute Deviation) Suppose we have data $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ related by model

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

The **least absolute deviation line** is obtained by minimizing

$$\min_{\beta_0, \beta_1} \sum_{i=1}^n |Y_i - \beta_0 - \beta_1 X_i|$$

Result 11.2.

$$\arg \min_m \sum_{i=1}^n |X_i - m| = \text{median}\{X_1, \dots, X_n\}$$

Result 11.3. The LAD line passes through a pair of points $(X_i, Y_i), (X_j, Y_j)$.

Proof. For simplicity, assume n odd.

Begin by fixing β_1 and defining

$$Z_i = Y_i - \beta_1 X_i$$

Then

$$\begin{aligned} \arg \min_{\beta_0} \sum_{i=1}^n |Z_i - \beta_0| &= \arg \min_{\beta_0} \sum_{i=1}^n |Z_i - \beta_0| \\ &= \text{median}\{Z_1, \dots, Z_n\} \\ &= Z_{i_0} \text{ for some } i_0 \\ &= Y_{i_0} - \beta_1 X_{i_0} \end{aligned}$$

This implies the LAD line for fixed β_1 passes through some point (X_{i_0}, Y_{i_0}) .

by shifting the coordinate system, we can assume that the LAD line passes through the origin

$$Y - Y_{i_0} = \beta_1 (X - X_{i_0})$$

We solve for β_1 with

$$\begin{aligned} &\min_{\beta_1} \sum_{i=1}^n |(Y_i - Y_{i_0}) - \beta_1 (X_i - X_{i_0})| \\ &= \min_{\beta_1} \sum_{i=1}^n |Y_i - \beta_1 X_i| \end{aligned}$$

For simplicity, we just write Y_i, X_i

$$\begin{aligned} &\min_{\beta_1} \sum_{i=1}^n |Y_i - \beta_1 X_i| \\ &= \min_{\beta_1} \sum_{i=1}^n |X_i| \left| \frac{Y_i}{X_i} - \beta_1 \right| \end{aligned}$$

This is a weighted median problem, the sum of absolute deviations is piecewise linear between each data point and convex. Hence, the minimum is attained at some i_* , i.e.

$$\hat{\beta}_1 = \frac{Y_{i_*}}{X_{i_*}}$$

The LAD line is

$$Y = Y_{i_0} + \frac{Y_{i_*} - Y_{i_0}}{X_{i_*} - X_{i_0}} (X - X_{i_0})$$

✓

Remark. The LAD line can be computed by checking over all the $\binom{n}{2}$ pairwise lines determined by the data points. These lines are called the **elemental lines**.

Remarks. Note that

1. The LAD line is one of the elemental lines
2. The LAD estimates are the MLEs where the errors have the Laplace / Double Exponential distribution

$$Ae^{\frac{-|x|}{B}}$$

3. The slope of the LS line is a weighted average of the slopes of the elemental lines.

12 Class 12

12.1 Confidence Intervals

Definition 12.1. For an unknown parameter θ and a sample $X_1, X_2, \dots, X_n$, A $100(1 - \alpha)\%$ confidence interval for θ is a random interval

$$[L(X_1, X_2, \dots, X_n), U(X_1, X_2, \dots, X_n)]$$

such that

$$P(L \leq \theta \leq U) = 1 - \alpha$$

Remark. Interpretation of CI: If the experiment is repeated, $100(1 - \alpha)\%$ of intervals will contain true θ .

12.1.1 Exact Confidence Intervals

Definition 12.2. (t-distribution). If $U \sim N(0, 1)$ and $V \sim \chi_d^2$ and U, V independent, then

$$\frac{U}{\sqrt{\frac{V}{d}}} \sim t_d$$

i.e. the t -distribution with d degrees of freedom.

Example. (CI for μ, σ^2 , normal data) Suppose $X_1, X_2, \dots, X_n \sim \text{iid } N(\mu, \sigma^2)$ where μ, σ^2 unknown.

Confidence interval for μ :

1. Estimate μ with $\hat{\mu} = \bar{X}$
2. Find distribution of estimate

$$\begin{aligned} \bar{X} &\sim N\left(\mu, \frac{\sigma^2}{n}\right) \\ \Rightarrow Z = \frac{\bar{X} - \mu}{\frac{\sigma}{\sqrt{n}}} &\sim N(0, 1) \\ \Rightarrow P\left(\left|\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}\right| \leq z_{\alpha/2}\right) &= 1 - \alpha \\ \Rightarrow P\left(|\bar{X} - \mu| \leq \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) &= 1 - \alpha \\ \Rightarrow P\left(-\frac{\sigma}{\sqrt{n}} z_{\alpha/2} \leq \mu \leq \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) &= 1 - \alpha \\ \Rightarrow P\left(\bar{X} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \leq \mu \leq \bar{X} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2}\right) &= 1 - \alpha \end{aligned}$$

3. Estimate σ^2 :

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \sim \frac{\sigma^2}{n-1} \chi_{n-1}^2$$

Then, define

$$\begin{aligned} \hat{Z} &:= \frac{\bar{X} - \mu}{s/\sqrt{n}} \\ &= \frac{\sqrt{n}(\bar{X} - \mu)/\sigma}{\sqrt{s^2/\sigma^2}} \\ &= \frac{N(0, 1)}{\sqrt{\chi_{n-1}^2/(n-1)}} \\ &\sim t_{n-1} \end{aligned}$$

Hence

$$\begin{aligned}
P(|\hat{Z}| \leq t_{n-1, \alpha/2}) &= 1 - \alpha \\
\Rightarrow P\left(-t_{n-1, \alpha/2} \leq \left|\frac{\bar{X} - \mu}{s/\sqrt{n}}\right| \leq t_{n-1, \alpha/2}\right) &= 1 - \alpha \\
\Rightarrow P\left(\bar{X} - t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} \leq \mu \leq \bar{X} + t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}\right)
\end{aligned}$$

Confidence interval for σ^2

1. Estimate σ^2 with

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

2. Find the distribution of the estimate

$$s^2 \sim \frac{\sigma^2}{n-1} \chi_{n-1}^2 \Rightarrow \frac{s^2}{\sigma^2} \sim \frac{\chi_{n-1}^2}{n-1}$$

3. Since the distribution is free of unknown parameters, we can immediately find a confidence interval. Define $\chi_{n-1, \alpha}$ be the $(1 - \alpha)$ -th percentile of the χ_{n-1}^2 distribution.

$$\begin{aligned}
P\left(\chi_{n-1, 1-\alpha/2}^2 \leq \frac{(n-1)s^2}{\sigma^2} \leq \chi_{n-1, \alpha/2}^2\right) &= 1 - \alpha \\
\Rightarrow P\left(\frac{(n-1)s^2}{\chi_{n-1, \alpha/2}^2} \leq \sigma^2 \leq \frac{(n-1)s^2}{\chi_{n-1, 1-\alpha/2}^2}\right) &= 1 - \alpha
\end{aligned}$$

12.1.2 Asymptotic Confidence Intervals

Example. (Confidence interval for μ , arbitrary distribution): Suppose $X_1, X_2, \dots, X_n \sim iid F$ with $\mathbb{E}[X_1] = \mu, Var(X_1) = \sigma^2$.

Confidence interval for μ :

1. By CLT

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$$

2. By LLN,

$$s^2 \xrightarrow{p} \sigma^2 \Rightarrow s \xrightarrow{p} \sigma$$

3. By Slutsky's

$$\frac{\sqrt{n}(\bar{X} - \mu)}{s} \xrightarrow{d} N(0, 1)$$

4. Find the CI

$$\begin{aligned}
P\left(\mu \in \left[\bar{X} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{s}{\sqrt{n}}\right]\right) &\xrightarrow{n \rightarrow \infty} 1 - \alpha \\
\Rightarrow \left[\bar{X} - z_{\alpha/2} \frac{s}{\sqrt{n}}, \bar{X} + z_{\alpha/2} \frac{s}{\sqrt{n}}\right] &\text{ is an asymptotically } 100(1 - \alpha) \% \text{ confidence interval for } \mu
\end{aligned}$$

Remark. CLT of estimate and consistent estimation of variance gives asymptotic confidence intervals.

Example. (Poisson distribution) Let $X_1, \dots, X_n \sim Pois(\lambda)$. To find CI for λ ,

1. Estimate λ by $\hat{\lambda} = \bar{X}$
2. By CLT

$$\sqrt{n}(\hat{\lambda} - \lambda) \xrightarrow{d} N(0, \lambda)$$

Hence

$$Z = \frac{\sqrt{n}(\hat{\lambda} - \lambda)}{\sqrt{\lambda}} \xrightarrow{d} N(0, 1)$$

3. Hence, by Slutsky's

$$\hat{Z} = \frac{\sqrt{n}(\hat{\lambda} - \lambda)}{\sqrt{\hat{\lambda}}} \xrightarrow{d} N(0, 1)$$

Hence,

$$P\left(-z_{\alpha/2} \leq \frac{\sqrt{n}(\hat{\lambda} - \lambda)}{\sqrt{\hat{\lambda}}} \leq z_{\alpha/2} \xrightarrow{n \rightarrow \infty} 1 - \lambda\right)$$

Definition 12.3. (Wald Interval) Suppose $\{f_\theta(x) : \theta \in \Omega\}$ is a parametric family of distributions, and $X_1, \dots, X_n$ iid from f_θ , the **Wald Interval** for θ is the asymptotically $100(1 - \alpha)\%$ CI for θ given by

$$\left[\hat{\theta} - \frac{z_{\alpha/2}}{\sqrt{nI(\hat{\theta})}}, \hat{\theta} + \frac{z_{\alpha/2}}{\sqrt{nI(\hat{\theta})}} \right]$$

where $\hat{\theta}$ is the MLE estimate.

Proof. Since

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} N\left(0, \frac{1}{I(\theta)}\right)$$

Estimate $I(\theta)$ by $I(\hat{\theta})$. If $I(\theta)$ continuous in θ by continuous mapping,

$$I(\hat{\theta}) \xrightarrow{p} I(\theta)$$

By Slutsky's

$$\sqrt{nI(\hat{\theta})}(\hat{\theta} - \theta) \xrightarrow{d} N(0, 1)$$

Hence

$$P\left(-z_{\alpha/2} \leq \sqrt{nI(\hat{\theta})}(\hat{\theta} - \theta) \leq z_{\alpha/2}\right) \xrightarrow{n \rightarrow \infty} 1 - \alpha$$

✓

13 Class 13

13.1 Proof of Weak of Law of Large Numbers without finite variance

Theorem 13.1. (Weak law of large numbers)

■ **Proof.**

✓

14 Class 14

14.1 Hypothesis testing

Remark. Suppose $X_1, \dots, X_n$ iid from $f_\theta(x)$, where $\theta \in \Omega$. We want to test the following hypotheses

1. Simple vs simple

$$H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta = \theta_1$$

2. Simple vs composite

$$\begin{aligned} H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta \neq \theta_1 \text{ OR} \\ H_1 : \theta < \theta_0 \\ H_1 : \theta > \theta_0 \end{aligned}$$

3. Composite vs composite

$$\begin{aligned} H_0 : \theta < \theta_0 \text{ vs } H_1 : \theta > \theta_1 \text{ OR} \\ H_0 : \theta > \theta_0 \text{ vs } H_1 : \theta < \theta_1 \text{ OR} \end{aligned}$$

Definition 14.1. (Test function) A test function ϕ is a function $\phi : (X_1, X_2, \dots, X_n) \rightarrow \{0, 1\}$ such that

$$\begin{aligned} \phi = 0 &\implies H_0 \text{ is accepted} \\ \phi = 1 &\implies H_0 \text{ is rejected} \end{aligned}$$

Definition 14.2. (Type I, Type II error) The two types of error in testing problems are

- Type I error: rejecting H_0 when H_0 true (false positive)
- Type II error: not rejecting H_0 when H_1 true (false negative)

Definition 14.3. (Level of a test): The significance level of a test is the probability of Type I error

$$\alpha = P_{H_0}(\text{reject } H_0) = P_{H_0}(\phi = 1) = \text{Type I error rate} = \text{False Positive Rate}$$

Definition 14.4. (p-value): The p-value is the smallest significance level at which the test rejects H_0 .

Definition 14.5. (Power): Let β be the probability of Type II error

$$\beta = P_{H_1}(\text{accept } H_0) = P_{H_1}(\phi = 0) = \text{Type II error rate} = \text{False Negative Rate}$$

The power of a test is

$$1 - \beta = P_{H_1}(\text{reject } H_0) = P_{H_1}(\phi = 1)$$

14.1.1 Likelihood Ratio Test and Neyman-Pearson Lemma

Definition 14.6. (MP test): The **most powerful (MP) test** is a test which maximizes power $(1 - \beta)$ at a given level (α) , i.e.

$$\begin{aligned} \max_{\phi} \mathbb{E}_{H_1} [\phi] \\ \text{subject to } \mathbb{E}_{H_0} [\phi] \leq \alpha \end{aligned}$$

Definition 14.7. (Likelihood ratio test): Reject H_0 for small values of the likelihood ratio

$$L(X_1, \dots, X_n) = \frac{f_{\theta_0}(X_1, \dots, X_n)}{f_{\theta_1}(X_1, \dots, X_n)}$$

Remark. "The points which give the strongest evidence in favor of H_1 over H_0 ". check what this means

Theorem 14.8. (Neyman-Pearson Lemma) Under a simple vs simple hypothesis test, for a constant $c > 0$, define the test function

$$\phi_0(X_1, X_2, \dots, X_n) = L(X_1, \dots, X_n) < c$$

where c is chosen such that

$$P(\text{Type I error}) = P_{H_0}(L(X_1, \dots, X_n)) = \alpha$$

Then ϕ_0 is a MP test for the testing problem

Proof. Let ϕ be any other test with significance level at most α . WTS that

$$\mathbb{E}_{H_1}[\phi_0] \geq \mathbb{E}_{H_1}[\phi]$$

Let

$$g(\mathbf{x}) = (\phi_0(\mathbf{x}) - \phi(\mathbf{x}))(cf_{\theta_1}(\mathbf{x}) - f_{\theta_0}(\mathbf{x}))$$

Note that

$$\phi_0(\mathbf{x}) > \phi(\mathbf{x}) \implies f_{\theta_0}(\mathbf{x}) < cf_{\theta_1}(\mathbf{x})$$

$$\phi_0(\mathbf{x}) < \phi(\mathbf{x}) \implies f_{\theta_0}(\mathbf{x}) \geq cf_{\theta_1}(\mathbf{x})$$

This means

$$\begin{aligned} g(\mathbf{x}) &\geq 0 \\ \implies \int g(\mathbf{x}) d\mathbf{x} &\geq 0 \\ \implies \int (\phi_0(\mathbf{x}) - \phi(\mathbf{x}))(cf_{\theta_1}(\mathbf{x}) - f_{\theta_0}(\mathbf{x})) d\mathbf{x} &\geq 0 \\ \implies c \int (\phi_0(\mathbf{x}) - \phi(\mathbf{x})) f_{\theta_1}(\mathbf{x}) d\mathbf{x} &\geq \int (\phi_0(\mathbf{x}) - \phi(\mathbf{x})) f_{\theta_0}(\mathbf{x}) d\mathbf{x} \\ \implies c(\mathbb{E}_{H_1}[\phi_0] - \mathbb{E}_{H_1}[\phi]) &\geq 0 \end{aligned}$$

Since

$$\begin{aligned} \int \phi_0(\mathbf{x}) f_{\theta_0}(\mathbf{x}) d\mathbf{x} &= \mathbb{E}_{H_0}[\phi_0] = \alpha \\ \int \phi(\mathbf{x}) f_{\theta_0}(\mathbf{x}) d\mathbf{x} &= \mathbb{E}_{H_0}[\phi] \leq \alpha \end{aligned}$$

Hence,

$$\mathbb{E}_{H_1}[\phi_0] \geq \mathbb{E}_{H_1}[\phi]$$

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Example. (Testing normal means) $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, σ^2 known. Want to test

$$H_0: \mu = \mu_0 \text{ vs } H_1: \mu = \mu_1 \neq \mu_0$$

The likelihood ratio is

$$\begin{aligned} L(X_1, \dots, X_n) &= \frac{\exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_0)^2\right\}}{\exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_1)^2\right\}} \\ &= \exp\left\{-\frac{1}{2\sigma^2} \left(\sum_{i=1}^n (X_i - \mu_0)^2 - \sum_{i=1}^n (X_i - \mu_1)^2\right)\right\} \\ &= \exp\left\{-\frac{1}{2\sigma^2} \left(2(\mu_1 - \mu_0) \sum_{i=1}^n X_i + \mu_0^2 - \mu_1^2\right)\right\} \\ &= \exp\left\{-\frac{(\mu_1 - \mu_0)}{(\sigma^2)} \sum_{i=1}^n X_i\right\} \exp\left\{-\frac{1}{2\sigma^2} (\mu_0^2 - \mu_1^2)\right\} \end{aligned}$$

Case 1: $\mu_1 > \mu_0$, then we reject for large values of $\sum_{i=1}^n X_i$.

Choose c such that

$$P_{H_0} \left(\sum_{i=1}^n X_i > c \right) = \alpha$$

Since under H_0

$$\sum_{i=1}^n X_i \sim N(n\mu_0, n\sigma^2)$$

therefore

$$Z = \frac{\sum_{i=1}^n X_i - n\mu_0}{\sqrt{n}\sigma} \sim N(0, 1)$$

Therefore we need to choose c such that

$$\begin{aligned} P_{H_0} \left(\frac{\sum_{i=1}^n X_i - n\mu_0}{\sqrt{n}\sigma} > \frac{c - n\mu_0}{\sqrt{n}\sigma} \right) &= \alpha \\ \Rightarrow P_{H_0} \left(N(0, 1) > \frac{c - n\mu_0}{\sqrt{n}\sigma} \right) &= \alpha \\ \Rightarrow \frac{c - n\mu_0}{\sqrt{n}\sigma} = z_\alpha \Rightarrow c &= n\mu_0 + z_\alpha \sigma \sqrt{n} \end{aligned}$$

Hence the MP test for when $\mu_1 > \mu_0$ rejects H_0 when

$$\sum_{i=1}^n X_i > n\mu_0 + z_\alpha \sigma \sqrt{n} \Leftrightarrow \frac{\sqrt{n}\bar{X} - \mu_0}{\sigma} > z_\alpha$$

Case 2: when $\mu_1 < \mu_0$, we reject for small values of $\bar{X}$. Choose c such that

$$P_{H_0}(\bar{X} < c) = \alpha$$

Therefore

$$P \left(\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} < \frac{\sqrt{n}(c - \mu_0)}{\sigma} \right) = \alpha$$

Hence

$$\frac{\sqrt{n}(c - \mu_0)}{\sigma} = -z_\alpha$$

Hence the MP test for when $\mu_1 < \mu_0$ rejects H_0 when

$$\frac{\sqrt{n}(\bar{X} - \mu_0)}{\sigma} < -z_\alpha$$

14.1.2 One sided composite hypothesis and UMP tests

Definition 14.9. (UMP test) Consider testing the following one-sided composite hypothesis

$$H_0 : \mu = \mu_0 \text{ vs } H_1 : \mu > \mu_0$$

The **uniformly most powerful test** is a test such that the power function is uniformly maximized for $\mu \neq \mu_0$ over all level α tests.

i.e. The UMP test is ϕ_0 such that

$$\mathbb{E}_{H_0} [\phi_0] = \alpha$$

and

$$\mathbb{E}_\mu [\phi_0] \geq \mathbb{E}_\mu [\phi]$$

for all $\mu \neq \mu_0$ and any ϕ such that $\mathbb{E}_{H_0} [\phi] = \alpha$

Remark. There exists no UMP test for two sided hypothesis because the UMP tests for 1 sided hypotheses have disjoint rejection regions.

15 Class 15

15.1 Generalized Likelihood Ratio Test

Definition 15.1. (GLRT, simple null) Let $\{f_\theta(x) : \theta \in \Omega\}$ be a parametric model and let $\theta_0 \in \Omega$ be a particular parameter value. Consider the testing problem

$$H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta \neq \theta_0$$

Then, the GLRT rejects H_0 for small values of

$$\Lambda = \frac{L(\theta_0 | X_1, \dots, X_n)}{\max_{\theta \in \Omega} L(\theta | X_1, \dots, X_n)}$$

Where

$$L(\theta | X_1, \dots, X_n) = \prod_{i=1}^n f_\theta(X_i)$$

is the likelihood ratio function.

Example. (Testing normal means for two sided hypothesis)

Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, σ^2 known, with test

$$H_0 : \mu = \mu_0 \text{ vs } H_1 : \mu \neq \mu_0$$

To compute the GLRT, note that

$$\max_{\mu} L(\mu | X_1, \dots, X_n) = L(\hat{\mu} | X_1, \dots, X_n)$$

where $\hat{\mu} = \bar{X}$ is the MLE for μ .

Then

$$\begin{aligned} \Lambda &= \frac{\exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_0)^2\right\}}{\exp\left\{-\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2\right\}} \\ &= \exp\left\{-\frac{1}{2\sigma^2} n (\bar{X} - \mu_0)^2\right\} \end{aligned}$$

We reject for small Λ , which is equivalent to large values of

$$-2 \log \Lambda = \frac{n (\bar{X} - \mu_0)^2}{\sigma^2}$$

Under H_0 ,

$$n \frac{(\bar{X} - \mu_0)^2}{\sigma^2} \sim \chi_1^2$$

Hence we reject H_0 when

$$\frac{(\bar{X} - \mu_0)^2}{\sigma^2} > \chi_{1,\alpha}^2 \Leftrightarrow \left| \frac{\sqrt{n} (\bar{X} - \mu_0)}{\sigma} \right| \leq z_{\alpha/2}$$

Example. (MVG) $X_1, \dots, X_n \sim N_p(\boldsymbol{\mu}, \Sigma)$ where Σ known, and test

$$H_0 : \boldsymbol{\mu} = \boldsymbol{\mu}_0 \text{ vs } H_1 : \boldsymbol{\mu} \neq \boldsymbol{\mu}_0$$

GLRT rejects when

$$-2 \log \Lambda = n (\bar{X} - \boldsymbol{\mu})^T \Sigma^{-1} (\bar{X} - \boldsymbol{\mu}) > \chi_{p,\alpha}^2$$

Example. (One sided hypothesis)

Recall $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ where σ^2 known,

$$H_0 : \mu = \mu_0 \text{ vs } H_1 : \mu > \mu_0$$

Reject H_0 when

$$\sqrt{n} \frac{\bar{X} - \mu_0}{\sigma} > z_\alpha$$

We claim that this is the GLRT.

To compute GLRT, we need to first maximize

$$\max_{\mu > \mu_0} L(\mu | X_1, \dots, X_n) = \max_{\mu > \mu_0} \left(\frac{1}{\sqrt{2\pi}\sigma} \right)^n \exp \left\{ -\frac{\sum_{i=1}^n (X_i - \mu)^2}{2\sigma^2} \right\}$$

This equivalently minimizes

$$\min_{\mu > \mu_0} \sum (X_i - \mu)^2 = \min_{\mu > \mu_0} \left(\sum_{i=1}^n X_i^2 - 2n\bar{X} + n\mu_0^2 \right)$$

Case 1: $\bar{X} > \mu_0$:

$$\min_{\mu > \mu_0} \sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \bar{X})^2$$

Case 2: $\bar{X} \leq \mu_0$:

$$\min_{\mu > \mu_0} \sum_{i=1}^n (X_i - \mu)^2 = \sum_{i=1}^n (X_i - \mu_0)^2$$

Hence

$$\begin{aligned} \Lambda &= \frac{\exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_0)^2 \right\}}{\exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \bar{X})^2 \right\} \mathbf{1}[\bar{X} > \mu_0] + \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_0)^2 \right\} \mathbf{1}[\bar{X} \leq \mu_0]} \\ &= \begin{cases} \exp \left\{ -\frac{n}{2\sigma^2} (\bar{X} - \mu_0)^2 \right\} & \text{if } \bar{X} > \mu_0 \\ 1 & \text{otherwise} \end{cases} \end{aligned}$$

Reject H_0 for small values of Λ , i.e. large values of $\bar{X} - \mu_0$. Hence the level α GLRT rejects H_0 when

$$\sqrt{n} \frac{\bar{X} - \mu_0}{\sigma} > z_\alpha$$

15.2 Asymptotic distribution of the GLRT

In general, the exact distribution of $-2 \log \Lambda$ under H_0 may not have a simple form, but it can be approximated by chi-squared when $n \rightarrow \infty$.

Theorem 15.2. Suppose $f_{\theta(x): \theta \in \mathbb{R}^k}$ is a parametric model indexed by k -dim parameter vector.

Let $X_1, \dots, X_n$ iid $f(x|\theta)$ and test

$$H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta \neq \theta_0$$

Under regularity conditions

$$-2 \log \Lambda \xrightarrow{d} \chi_k^2$$

■ **Proof.** Proof for case $k = 1$

✓

15.3 GLRT for submodels

Suppose $\Omega_0 \subset \Omega$ is a lower dimension subspace of parameter space $\Omega = \mathbb{R}^k$. We want to test

$$H_0 : \theta \in \Omega_0 \text{ vs } H_1 : \theta \notin \Omega_0$$

The GLRT is

$$\Lambda = \frac{\max_{\theta \in \Omega_0} L(\theta | X_1, \dots, X_n)}{\max_{\theta \in \Omega} L(\theta | X_1, \dots, X_n)}$$

Theorem 15.3. Let $\{f_\theta : \theta \in \Omega\}$ be a parametric model, and let $X_1, \dots, X_n$ be iid $f_{\theta_0}(x)$. Suppose θ_0 is an interior

point of both Ω_0 and Ω , then under regularity conditions,

$$-2 \log \Lambda \xrightarrow{d} \chi_d^2$$

Where

$$d = \dim \Omega - \dim \Omega_0$$

Example. (Testing normal means, unknown variance)

Suppose $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, where σ^2 unknown. Test

$$H_0 : \mu = \mu_0 \text{ vs } H_1 : \mu \neq \mu_0$$

Then

$$\Lambda = \frac{\max_{\sigma^2} \left(\frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu_0)^2 \right\} \right)}{\max_{\mu, \sigma^2} \left(\frac{1}{\sqrt{2\pi}\sigma} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (X_i - \mu)^2 \right\} \right)}$$

The denominator is maximized at

$$\hat{\mu} = \bar{X}, \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2$$

The numerator is maximized by

$$\hat{\sigma}_{H_0}^2 = \sum_{i=1}^n (X_i - \mu_0)^2$$

Therefore

$$\begin{aligned} \Lambda &= \left(\frac{\hat{\sigma}^2}{\hat{\sigma}_{H_0}^2} \right)^{\frac{n}{2}} \\ &= \left(\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \mu_0)^2} \right)^{\frac{n}{2}} \\ &= \left(\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sum_{i=1}^n (X_i - \bar{X})^2 + n(\bar{X} - \mu_0)^2} \right)^{\frac{n}{2}} \end{aligned}$$

Since

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Λ small is equivalent to large values of

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right|$$

Hence *GLRT* rejects H_0 when

$$-2 \log \Lambda = n \log \left(1 + \frac{n(\bar{X} - \mu_0)^2}{\sum_{i=1}^n (X_i - \bar{X})^2} \right) \approx \frac{n(\bar{X} - \mu_0)^2}{\sum_{i=1}^n (X_i - \bar{X})^2}$$

Since $\log(1+x) \approx x$.

Since

$$n(\bar{X} - \mu_0)^2 \xrightarrow{d} \sigma^2 \chi_1^2$$

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \xrightarrow{p} \sigma^2$$

Therefore

$$-2 \log \Lambda \xrightarrow{d} \chi_1^2$$

16 Lecture 16

16.1 GLRT for submodels, cont'd

Example. (Hardy-Weinberg Equilibrium) An individual has genotype AA, Aa, aa . Draw n random observations and denote

$$N_{AA}, N_{Aa}, N_{aa}$$

such that

$$N_{AA} + N_{Aa} + N_{aa} = n$$

We model this with multinomial distribution with parameters p_{AA}, p_{Aa}, p_{aa} where

$$p_{AA} + p_{Aa} + p_{aa} = 1$$

Then

$$(N_{AA}, N_{Aa}, N_{aa}) \sim \text{Multi}(n, p_{AA}, p_{Aa}, p_{aa})$$

This has PMF

$$P(N_1 = n_1, N_2 = n_2, N_3 = n_3) = \frac{n! p_1^{n_1} p_2^{n_2} p_3^{n_3}}{n_1! n_2! n_3!}$$

We want to test

$$H_0 : p_1 = \theta^2, p_2 = 2\theta(1 - \theta), p_3 = (1 - \theta)^2$$

vs

$$H_1 : H_0 \text{ not true}$$

Here, the reduced model is

$$\Omega_0 = \{\theta \in (0, 1), p_1 = \theta^2, p_2 = 2\theta(1 - \theta), p_3 = (1 - \theta)^2\}$$

and

$$\Omega = \{(p_1, p_2, p_3) : p_1 + p_2 + p_3 = 1\}$$

Note

$$\dim \Omega_0 = 1, \dim \Omega = 2$$

Under full model (H_1), denote

$$\hat{p}_1, \hat{p}_2, \hat{p}_3$$

Under submodel, (H_0), denote

$$\hat{p}_{1,H_0}, \hat{p}_{2,H_0}, \hat{p}_{3,H_0}$$

1. Solve MLE for full model

$$\hat{p}_1 = \frac{n_1}{n}, \hat{p}_2 = \frac{n_2}{n}, \hat{p}_3 = \frac{n_3}{n}$$

2. Solve MLE for submodel

$$\begin{aligned} l(\theta) &\propto (\theta^2)^{n_1} (2\theta(1 - \theta))^{n_2} (1 - \theta)^{2n_3} \\ &\propto \theta^{2n_1 + n_2} (1 - \theta)^{n_2 + 2n_3} \end{aligned}$$

Setting $\frac{\partial}{\partial \theta} \log l(\theta) = 0$,

$$\begin{aligned} \hat{\theta} &= \frac{2n_1 + n_2}{2n_1 + 2n_2 + 2n_3} \\ &= \frac{2n_1 + n_2}{2n} \end{aligned}$$

Then,

$$\Lambda = \left(\frac{\hat{p}_{1,H_0}}{\hat{p}_1} \right)^{n_1} \left(\frac{\hat{p}_{2,H_0}}{\hat{p}_2} \right)^{n_2} \left(\frac{\hat{p}_{3,H_0}}{\hat{p}_3} \right)^{n_3}$$

and under H_0

$$-2 \log \Lambda = 2n_1 \log \left(\frac{\hat{p}_1}{\hat{p}_{1,H_0}} \right) + 2n_2 \log \left(\frac{\hat{p}_2}{\hat{p}_{2,H_0}} \right) + 2n_3 \log \left(\frac{\hat{p}_3}{\hat{p}_{3,H_0}} \right) \xrightarrow{d} \chi_1^2$$

Example. (Two sample t-test, equal variances) Suppose $X_1, \dots, X_m \sim N(\mu_1, \sigma^2)$, $Y_1, \dots, Y_n \sim N(\mu_2, \sigma^2)$, σ^2 unknown. Test

$$H_0 : \mu_1 = \mu_2 \text{ vs } H_1 : \mu_1 \neq \mu_2$$

The likelihood function is

$$L(\mu_1, \mu_2, \sigma^2 | X_1, \dots, X_m, Y_1, \dots, Y_n) = \left(\frac{1}{\sqrt{2\pi\sigma}} \right)^m \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^m (X_i - \mu_1)^2 \right\} \left(\frac{1}{\sqrt{2\pi\sigma}} \right)^n \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (Y_i - \mu_2)^2 \right\}$$

The GLRT is based on

$$\Lambda = \frac{\max_{\mu, \sigma^2} L(\mu, \mu, \sigma^2 | \mathbf{X}, \mathbf{Y})}{\max_{\mu_1, \mu_2, \sigma^2} L(\mu_1, \mu_2, \sigma^2 | \mathbf{X}, \mathbf{Y})}$$

Rejecting H_0 for small values of Λ is equivalent to rejecting for large values of

$$\left| \frac{\bar{X} - \bar{Y}}{s_P \sqrt{\frac{1}{m} + \frac{1}{n}}} \right|$$

Where s_p^2 is the pooled sample variance

$$s_p^2 = \frac{1}{n+m-2} \left\{ \sum_{i=1}^m (X_i - \bar{X})^2 + \sum_{j=1}^n (Y_j - \bar{Y})^2 \right\}$$

Note that

$$\begin{aligned} \sum_{i=1}^m (X_i - \bar{X})^2 &\sim \sigma \chi_{m-1}^2 \\ \sum_{j=1}^n (Y_j - \bar{Y})^2 &\sim \sigma \chi_{n-1}^2 \end{aligned}$$

And the following are independent.

$$\bar{X}, \bar{Y}, \sum_{i=1}^m (X_i - \bar{X})^2, \sum_{j=1}^n (Y_j - \bar{Y})^2$$

Under H_0 ,

$$\frac{\bar{X} - \bar{Y}}{\sqrt{\frac{\sigma^2}{m} + \frac{\sigma^2}{n}}} \sim N(0, 1)$$

and

$$\frac{s_p^2}{\sigma^2} \sim \frac{\chi_{m+n-2}^2}{m+n-2}$$

Hence

$$T = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2}$$

The GLRT level α test rejects when

$$|T| \geq t_{m+n-2, \alpha/2}$$

Example. (Two sample t-test, unequal variances) Suppose $X_1, \dots, X_m$ iid $N(\mu_1, \sigma_1^2)$, and $Y_1, Y_2, \dots, Y_n$ iid $N(\mu_2, \sigma^2)$.

Consider the hypothesis

$$H_0 : \mu_1 = \mu_2 \text{ vs } H_1 : \mu_1 \neq \mu_2$$

This is the **Behrens-Fisher problem**.

The **GLRT** does not have a simple closed form.

A natural statistic is

$$T_{\text{unequal}} = \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}}$$

Where

$$\begin{aligned} s_1^2 &= \frac{1}{m} \sum_{i=1}^m (X_i - \bar{X})^2 \\ s_2^2 &= \frac{1}{n} \sum_{j=1}^n (Y_j - \bar{Y})^2 \end{aligned}$$

The distribution of T_{unequal} is no longer exactly t -distribution, and instead depends on σ_1^2/σ_2^2 .

Welch t-test

The distribution of T_{unequal} is approximated by a t -distribution with γ degrees of freedom, where

$$\min\{m, n\} \leq \gamma \leq n + m - 2$$

Hence, the test which rejects H_0 when

$$\left| \frac{\bar{X} - \bar{Y}}{\sqrt{\frac{s_1^2}{m} + \frac{s_2^2}{n}}} \right| > t_{\gamma, \frac{\alpha}{2}}$$

is an approximately level α test.

Proof: When sample sizes are large, i.e.

$$\min\{m, n\} \rightarrow \infty$$

Then

$$T_{\text{unequal}} \xrightarrow{d} N(0, 1)$$

Hence the test which rejects H_0 when

$$|T_{\text{unequal}}| > z_{\alpha/2}$$

is asymptotically level α .

17 Class 17

17.1 Multiple linear regression

17.1.1 Setup

Under MLR, assume the outcome variable y_i is linearly related to p covariates, with n samples, we have

$$y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_p x_{pi} + \epsilon_i$$

Where the error variables are assumed

$$\epsilon_i \sim N(0, \sigma^2)$$

i.e. error terms are assumed to have mean 0, and are independently and normally distributed with identical variances.

In matrix form, we write

$$\begin{aligned} \mathbf{Y} &= \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \\ \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} &= \begin{pmatrix} 1 & x_{11} & \dots & x_{p1} \\ 2 & x_{12} & \dots & x_{p2} \\ \vdots & \vdots & \ddots & \vdots \\ n & x_{1n} & \dots & x_{pn} \end{pmatrix} \begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{pmatrix} + \begin{pmatrix} \epsilon_0 \\ \epsilon_1 \\ \vdots \\ \epsilon_p \end{pmatrix} \end{aligned}$$

i.e.

- $\mathbf{Y} \in \mathbb{R}^{n \times 1}$
- $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$
- $\boldsymbol{\beta} \in \mathbb{R}^{p+1}$
- $\boldsymbol{\epsilon} \in \mathbb{R}^{n \times 1}$

17.1.2 Least Square Estimate

Definition 17.1. (Least square estimate of MLR): Suppose

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I})$$

The least square estimate of $\boldsymbol{\beta}$, denoted $\hat{\boldsymbol{\beta}}$ is defined as

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= \arg \min_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 \\ &= \arg \min_{\boldsymbol{\beta}} \sum_{i=1}^n (y_i - \mathbf{x}_i^\top \boldsymbol{\beta})^2 \end{aligned}$$

Where

$$\mathbf{x}_i = \begin{pmatrix} 1 \\ x_{1i} \\ x_{2i} \\ \vdots \\ x_{pi} \end{pmatrix}$$

Result 17.2. If $\text{rank}(\mathbf{X}) = p + 1$, then the closed form solution for $\hat{\boldsymbol{\beta}}$ is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

Proof. We can express the sum of squared errors as

$$\begin{aligned} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 &= (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}) \\ &= \mathbf{Y}^\top \mathbf{Y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{Y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} \end{aligned}$$

Taking the derivative with respect to $\boldsymbol{\beta}$ gives the gradient vector

$$\nabla_{\boldsymbol{\beta}} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = -2\mathbf{X}^\top \mathbf{Y} + 2\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$$

Setting this equal to $\mathbf{0}$, we get the **normal equation**

$$\mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{X}^\top \mathbf{Y}$$

Since $\mathbf{X}$ has full column rank (i.e. $\text{rank } p + 1$), $\text{rank}(\mathbf{X}^\top \mathbf{X}) = p + 1$, hence

$$\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

✓

17.1.3 Properties of the LS estimate

Result 17.3. The LS estimate is unbiased

$$\begin{aligned} \mathbb{E}[\hat{\beta}] &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbb{E}[\mathbf{Y}] \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \beta \\ &= \beta \end{aligned}$$

Result 17.4. Variance of $\hat{\beta}$

$$\begin{aligned} \text{Var}(\hat{\beta}) &= \text{Var}(\mathbf{A}\mathbf{Y}) \\ &= \mathbf{A} \text{Var}(\mathbf{Y}) \mathbf{A}^\top \\ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \sigma^2 \mathbf{I} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \end{aligned}$$

Result 17.5. The LS estimate is normally distributed.

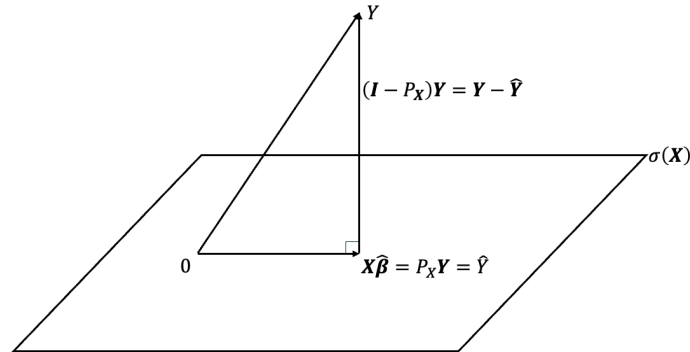
We know that

$$\mathbf{Y} \sim N_n(\mathbf{X}\beta, \sigma^2 \mathbf{I})$$

Hence $\hat{\beta} = \mathbf{A}\mathbf{Y}$, $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ is a linear combination of normals, hence

$$\hat{\beta} \sim N_{p+1}(\beta, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$$

17.1.4 Geometric interpretations of LS estimates



Remark. Any vector $\mathbf{v} = \mathbf{X}\beta$ is a linear combination of the columns of $\mathbf{X}$. Let $\sigma(\mathbf{X})$ denote the column space of $\mathbf{X}$.

The LS estimate is obtained by finding $\mathbf{X}\hat{\beta} \in \sigma(\mathbf{X})$ that is the closest Euclidean distance to $\mathbf{Y} \in \mathbb{R}^n$.
The predicted value is

$$\begin{aligned} \hat{\mathbf{Y}} &= \mathbf{X}\hat{\beta} \\ &= \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y} \\ &= \mathbf{P}_\mathbf{X} \mathbf{Y} \end{aligned}$$

Result 17.6. Properties of the projection matrix $\mathbf{P}_\mathbf{X}$

1. For any $\mathbf{v} \in \mathbb{R}^n$, $\mathbf{P}_\mathbf{X} \mathbf{v} \in \sigma(\mathbf{X})$
2. If $\mathbf{v} \in \sigma(\mathbf{X})$, $\mathbf{P}_\mathbf{X} \mathbf{v} = \mathbf{v}$

3. P_X is idempotent,

$$\begin{aligned} P_X^2 &= X (X^\top X)^{-1} X^\top X (X^\top X)^{-1} X^\top \\ &= P_X \end{aligned}$$

Definition 17.7. (Residuals): Define the residuals as

$$\begin{aligned} \mathbf{E} &= \begin{pmatrix} E_1 \\ E_2 \\ \dots \\ E_n \end{pmatrix} \\ &= \mathbf{Y} - \hat{\mathbf{Y}} \\ &= \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}} \\ &= (\mathbf{I} - \mathbf{P}_X)\mathbf{Y} \end{aligned}$$

Result 17.8. The predicted value $\hat{\mathbf{Y}}$ is orthogonal to the residual $\mathbf{E}$.

$$\begin{aligned} \mathbf{E}^\top \hat{\mathbf{Y}} &= \mathbf{Y}^\top (\mathbf{I} - \mathbf{P}_X) \mathbf{P}_X \mathbf{Y} \\ &= \mathbf{0} \end{aligned}$$

Hence by Pythagoras theorem

$$\begin{aligned} \|\mathbf{Y}\|^2 &= \|\hat{\mathbf{Y}}\|^2 + \|\mathbf{E}\|^2 \\ &= \|\mathbf{P}_X \mathbf{Y}\|^2 + \|(\mathbf{I} - \mathbf{P}_X) \mathbf{Y}\|^2 \end{aligned}$$

18 Class 18

18.1 Distribution of residuals

Result 18.1. Residuals are normally distributed.

$$\mathbf{E} \sim N_n(\mathbf{0}, \sigma^2 (\mathbf{I} - \mathbf{P}_X))$$

Proof. Normality: Since $\mathbf{Y} \sim N_n(\mathbf{X}\beta, \sigma^2 \mathbf{I})$, and $\mathbf{E} = (\mathbf{I} - \mathbf{P}_X)\mathbf{Y}$, $\mathbf{E}$ is a linear combination of normals.

Zero mean:

$$\begin{aligned}\mathbb{E}[\mathbf{E}] &= \mathbb{E}[\mathbf{Y} - \hat{\mathbf{Y}}] \\ &= \mathbb{E}[(\mathbf{I} - \mathbf{P}_X)\mathbf{Y}] \\ &= (\mathbf{I} - \mathbf{P}_X) \mathbb{E}[\mathbf{Y}] \\ &= (\mathbf{I} - \mathbf{P}_X) \mathbf{X}\beta \\ &= \mathbf{X}\beta - \mathbf{P}_X \mathbf{X}\beta \\ &= \mathbf{0}\end{aligned}$$

Variance:

$$\begin{aligned}\text{Var}(\mathbf{E}) &= \text{Var}((\mathbf{I} - \mathbf{P}_X)\mathbf{Y}) \\ &= \sigma^2 (\mathbf{I} - \mathbf{P}_X)^2 \\ &= \sigma^2 (\mathbf{I} - \mathbf{P}_X)\end{aligned}$$

✓

18.1.1 Estimating σ^2 in MLR

Definition 18.2. (Sum of square error, SSE): The **SSE** is defined as

$$\mathbf{E}^\top \mathbf{E} = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Result 18.3. The SSE follows a *chi-squared* distribution with $n - (p + 1)$ degrees of freedom.

$$\mathbf{E}^\top \mathbf{E} \sim \sigma^2 \chi_{n-(p+1)}^2$$

Result 18.4. An unbiased estimate of σ^2 is

$$\hat{\sigma}^2 = \frac{SSE}{n - (p + 1)} = \frac{\mathbf{E}^\top \mathbf{E}}{n - (p + 1)}$$

Result 18.5. Facts about the eigenvalues of $\mathbf{I} - \mathbf{P}_X$

1. $\text{rank}(\mathbf{X}) = p + 1 \implies \text{rank}(\mathbf{P}_X) = p + 1$
2. Since $(\mathbf{I} - \mathbf{P}_X)$ idempotent, it has eigenvalues of 0 or 1
3. $\mathbf{P}_X$ has $p + 1$ eigenvalues 1 and $n - (p + 1)$ eigenvalues 0
4. $(\mathbf{I} - \mathbf{P}_X)$ has $n - (p + 1)$ eigenvalues 1 and $p + 1$ eigenvalues 0.

Proof. We express $\mathbf{E} = (\mathbf{I} - \mathbf{P}_X)^{\frac{1}{2}} \boldsymbol{\eta}$, where $\boldsymbol{\eta} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$. Then

$$\mathbf{E}^\top \mathbf{E} = \boldsymbol{\eta}^\top (\mathbf{I} - \mathbf{P}_X) \boldsymbol{\eta}$$

By spectral theorem,

$$\mathbf{I} - \mathbf{P}_X = \mathbf{U} \mathbf{D} \mathbf{U}^\top$$

Where

$$\mathbf{U} \mathbf{U}^\top = \mathbf{U}^\top \mathbf{U} = \mathbf{I}$$

And

$$D = \begin{pmatrix} \mathbf{I}_{n-(p+1) \times n-(p+1)} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$$

Also note that

$$\mathbf{U}^\top \boldsymbol{\eta} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{U}^\top \mathbf{U}) = N_n(\mathbf{0}, \sigma^2 \mathbf{I})$$

Hence

$$\begin{aligned} \mathbf{E}^\top \mathbf{E} &= \boldsymbol{\eta}^\top (\mathbf{I} - \mathbf{P}_X) \boldsymbol{\eta} \\ &= \boldsymbol{\eta}^\top \mathbf{U} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \mathbf{U}^\top \boldsymbol{\eta} \\ &= \boldsymbol{\theta}^\top \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix} \boldsymbol{\theta} \\ &= \sum_{i=1}^{n-(p+1)} \theta_i^2 \\ &\sim \sigma^2 \chi_{n-(p+1)}^2 \end{aligned}$$

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18.2 Confidence interval and hypothesis testing for gression coefficients

Result 18.6. The residuals $\mathbf{E}$ and $\hat{\boldsymbol{\beta}}$ are independent.

Proof. Since $(\hat{\boldsymbol{\beta}}, \mathbf{E})$ normally distributed, suffices to show that

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{E}) = \mathbf{0}$$

Since

$$\begin{aligned} \hat{\boldsymbol{\beta}} &= \mathbf{A}\mathbf{Y}, \mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ \mathbf{E} &= \mathbf{B}\mathbf{Y}, \mathbf{B} = \mathbf{I} - \mathbf{P}_X \end{aligned}$$

$$\begin{aligned} \text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Y}) &= \sigma^2 \mathbf{A}\mathbf{B} \\ &= \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{I} - \mathbf{P}_X) \\ &= \sigma^2 \left((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \right) \\ &= \sigma^2 \left((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \right) \\ &= \mathbf{0} \end{aligned}$$

✓

Result 18.7. Since $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$ indepenent, where

$$\hat{\boldsymbol{\beta}} \sim N_{p+1}(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$$

and

$$\hat{\sigma}^2 = \frac{SSE}{n - (p + 1)} \sim \frac{\sigma^2 \chi_{n-(p+1)}^2}{n - (p + 1)}$$

Therefore, for $0 \leq i \leq p$

$$\frac{\hat{\beta}_i - \beta_i}{\hat{\sigma} \sqrt{(\mathbf{X}^\top \mathbf{X})_{ii}^{-1}}} \sim t_{n-(p+1)}$$

Confidence interval for β_i The following is a $100(1 - \alpha)\%$ confidence interval for β_i

$$\left[\hat{\beta}_i \pm t_{n-(p+1), \frac{\alpha}{2}} \hat{\sigma} \sqrt{(\mathbf{X}^\top \mathbf{X})_{ii}^{-1}} \right]$$

Hypothesis testing: to test

$$H_0 : \beta_i = 0 \text{ vs } H_1 : \beta_i \neq 0$$

at level α ,

We reject H_0 when

$$\left| \frac{\hat{\beta}_i}{\hat{\sigma} \sqrt{(\mathbf{X}^\top \mathbf{X})_{ii}^{-1}}} \right| > t_{n-(p+1), \frac{\alpha}{2}}$$

18.2.1 F-test and ANOVA

The MLR model is

$$y_i = \beta_0 + \beta_1 x_{1i} + \dots \beta_p x_{pi} + \epsilon_i$$

How well does the least squares model fit the data?

We test

$$H_0 : \beta_1 = \beta_2 = \dots = \beta_p = 0 \text{ vs } H_1 : \beta_i \neq 0 \text{ for some } 1 \leq i \leq p$$

Under H_0 , the model is

$$y_i = \beta_0 + \epsilon_i$$

Hence

$$\hat{\beta}_0 = \bar{Y}$$

Definition 18.8. (Sum of squares total, SST): The **SST** is defined as

$$\begin{aligned} SST &= \sum_{i=1}^n (y_i - \bar{Y})^2 \\ &= \mathbf{Y}^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{Y} \end{aligned}$$

Where $\mathbf{P}_1$ is the projection matrix on to the column space of $\mathbf{1}$

$$\mathbf{P}_1 = \mathbf{1} (\mathbf{1}^\top \mathbf{1})^{-1} \mathbf{1}^\top = \frac{\mathbf{1} \mathbf{1}^\top}{n}$$

Definition 18.9. (Multiple R^2 , coefficient of determinant): The R^2 or coefficient of determinant is defined as the proportion by which the LS model reduces the total sum of squares

$$\begin{aligned} R^2 &= 1 - \frac{SSE}{SST} \\ &= \frac{SST - SSE}{SST} \end{aligned}$$

Definition 18.10. (Adjusted R^2): The adjusted R^2 accounts for the fact that R^2 increases as more variables are added to the model. The adjusted R^2 is denoted

$$\bar{R}^2 = 1 - \frac{SSE/(n - (p + 1))}{SST/(n - 1)} = 1 - \frac{SSE}{SST} \frac{n - 1}{n - (p + 1)}$$

Definition 18.11. (SSR, Sum of squares regression): The **SSR** is defined as

$$\begin{aligned} SSR &= SST - SSE \\ &= \mathbf{Y}^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{Y} - \mathbf{Y}^\top (\mathbf{I} - \mathbf{P}_X) \mathbf{Y} \\ &= \mathbf{Y}^\top (\mathbf{P}_X - \mathbf{P}_1) \mathbf{Y} \end{aligned}$$

Result 18.12. Under H_0 , SSR follows a *chi-squared* distribution and is independent of SSE.

$$SSR \stackrel{H_0}{\sim} \sigma^2 \chi_p^2$$

Therefore

$$F = \frac{SSR/p}{SSE/n - (p + 1)} \stackrel{H_0}{\sim} F_{p, n-(p+1)}$$

19 Class 19

19.1 Comparison of confidence intervals

Remark. (Normal data): $X_1, \dots, X_n \sim N(\mu, \sigma^2)$

Parameter	σ^2 known	σ^2 unknown	Type
μ	$\left[\bar{X} \pm z_{1-\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$	$\left[\bar{X} \pm t_{n-1, 1-\alpha/2} \frac{S}{\sqrt{n}} \right]$	Exact
σ^2	NA	$\left[\frac{(n-1)S^2}{\chi_{n-1, 1-\alpha/2}^2}, \frac{(n-1)S^2}{\chi_{n-1, \alpha/2}^2} \right]$	Exact

Remark. (Arbitrary data): $X_1, \dots, X_n \sim F$ with $\mathbb{E}[X_1] = \mu, \text{Var}(X) = \sigma^2$ unknown

interval	$\left[\bar{X} \pm t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} \right]$	$\left[\bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}} \right]$
type of interval	Exact	Asymptotic
requires normality of data?	Yes	No

19.2 Confidence set

Definition 19.1. (Confidence set): A confidence set is a set A which is a function of $X_1, \dots, X_n$, such that

$$P(\mu \in A(X_1, \dots, X_n)) \geq 1 - \alpha$$

Result 19.2. The multivariate sample mean is multivariate normal

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n \bar{X}_i \sim N_p\left(\mu, \frac{\Sigma}{n}\right)$$

Example. When $p = 1$,

$$\bar{X} \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right) \Rightarrow \sqrt{n} \frac{\bar{X} - \mu}{\sigma} \sim \mathcal{N}(0, 1).$$

When: $p \geq 2$

Define

$$Z = \sqrt{n} \Sigma^{-1/2} (\bar{X} - \mu) \sim \mathcal{N}_p(0, I_p).$$

Then

$$Z^\top Z = \sum_{i=1}^p Z_i^2 = n(\bar{X} - \mu)^\top \Sigma^{-1} (\bar{X} - \mu) \sim \chi_p^2.$$

Hence,

$$P(Z^\top Z \leq \chi_{p, 1-\alpha}^2) = 1 - \alpha,$$

or equivalently,

$$P(n(\bar{X} - \mu)^\top \Sigma^{-1} (\bar{X} - \mu) \leq \chi_{p, 1-\alpha}^2) = 1 - \alpha.$$

Confidence set for μ

If Σ known

$$\{\mu \in \mathbb{R}^p : n(\bar{X} - \mu)^\top \Sigma^{-1} (\bar{X} - \mu) \leq \chi_{p, \alpha}^2\}$$

is a $100(1 - \alpha)\%$ confidence set for μ .

For the case where $p = 2, \Sigma = I$, the confidence set is

$$\{\mu \in \mathbb{R}^2 : (\bar{X} - \mu)^\top (\bar{X} - \mu) \leq \frac{\chi_{p, \alpha}^2}{n}\}$$

which describes a circle centered at $\bar{X}$ and radius

$$\sqrt{\frac{\chi_{p, \alpha}^2}{n}}$$

For $p = 2, \sigma \neq I$

$$\{\mu \in \mathbb{R}^2 : (\bar{X} - \mu)^\top \Sigma^{-1} (\bar{X} - \mu) \leq \frac{\chi_{p, \alpha}^2}{n}\}$$

If Σ unknown, By multivariate CLT

$$\sqrt{n}(\bar{X} - \mu) \xrightarrow{d} N_p(0, \Sigma)$$

Therefore,

$$Z = \sqrt{n}\Sigma^{-1}(\bar{\mathbf{X}} - \mu) \xrightarrow{d} N_p(0, 1)$$

To estimate Σ , we use the sample covariance matrix

$$S = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^T$$

$$S \xrightarrow{p} \Sigma$$

By multivariate Slutsky's

$$\tilde{Z} = \sqrt{n}S^{-\frac{1}{2}}(\bar{\mathbf{X}} - \mu) \xrightarrow{d} N_p(\mathbf{0}, I)$$

$$\tilde{Z}^T \tilde{Z} = n(\bar{\mathbf{X}} - \mu)^T S^{-1}(\bar{\mathbf{X}} - \mu)$$

$$\xrightarrow{d} \chi_p^2$$

Hence,

$$\{\mu \in \mathbb{R}^p : n(\bar{\mathbf{X}} - \mu)^T S^{-1}(\bar{\mathbf{X}} - \mu) \leq \chi_{p,\alpha}^2\}$$

is an asymptotically $100(1 - \alpha)\%$ confidence set of μ .

19.3 Hypothesis tests

19.3.1 Simple vs simple

Recall that for simple vs simple hypothesis, the most powerful test is given by the Neyman-Pearson lemma, where we reject for large values of

$$\Lambda(X_1, \dots, X_n) = \frac{f_{\theta_0}(X_1, \dots, X_n)}{f_{\theta_1}(X_1, \dots, X_n)}$$

19.3.2 Simple vs composite

Example. Recall that for testing normal means, for $X_1, \dots, X_n \sim N(\mu, \sigma^2)$, σ^2 known, the following are UMP tests

H_0	H_1	UMP Test
$H_0 : \mu = \mu_0$	$H_1 : \mu > \mu_0$	Reject $\bar{X} > \mu_0 + \frac{Z_{\alpha}}{\sqrt{n}}$
$H_0 : \mu = \mu_0$	$H_1 : \mu < \mu_0$	Reject $\bar{X} < \mu_0 - \frac{Z_{\alpha}}{\sqrt{n}}$
$H_0 : \mu = \mu_0$	$H_1 : \mu \neq \mu_0$	No UMP Test

For the last case, we can use the GLRT. The GLRT says to reject if

$$\left| \sqrt{n} \frac{(\bar{X} - \mu_0)}{\sigma} \right| > z_{\alpha/2}$$

Remark. Recall that for GLRT, testing

$$H_0 : \theta = \theta_0 \text{ vs } H_1 : \theta \neq \theta_0$$

$$\Lambda = \frac{L(\theta_0 | X_1, \dots, X_n)}{\max_{\theta \in \Omega} L(\theta | X_1, \dots, X_n)}$$

Under H_0 :

$$-2 \log \Lambda \xrightarrow{d} \chi_k^2$$

Remark. Recall GLRT for submodels, testing

$$H_0 : \theta \in \Omega_0 \subset \Omega \text{ vs } H_1 : \theta \notin \Omega_0$$

$$\Lambda = \frac{\max_{\theta \in \Omega_0} L(\theta | X_1, \dots, X_n)}{\max_{\theta \in \Omega} L(\theta | X_1, \dots, X_n)}$$

Under H_0 :

$$-2 \log \Lambda \xrightarrow{d} \chi_d^2$$

where

$$d = \dim \Omega - \dim \Omega_0$$

19.3.3 Composite vs composite

Remark. For composite vs composite tests, the GLRT test of level α rejects when

$$\left| \frac{\sqrt{n}(\bar{X} - \mu_0)}{S} \right| > t_{n-1, \frac{\alpha}{2}}$$

19.3.4 Asymptotic power of tests

Definition 19.3. (Consistent test) A level α test with power converging to 1 is known as a **consistent** test.

Example. $X_1, \dots, X_n \sim N(\theta, 1)$, test

$$H_0 : \theta = 0 \text{ vs } H_1 : \theta = \theta_1 > 0$$

The z -test rejects H_0 when

$$\sqrt{n}\bar{X} > z_\alpha$$

Under H_1 :

$$X_1, \dots, X_n \sim N(\theta_1, 1) \implies \sqrt{n}\bar{X} \sim N(\sqrt{n}\theta_1, 1)$$

The power of this test is

$$\begin{aligned} \text{Power}(\theta_1) &= P_{H_1}(\sqrt{n}\bar{X} > z_\alpha) \\ &= P(\sqrt{n}\bar{X} - \sqrt{n}\theta_1 > z_\alpha - \sqrt{n}\theta_1) \\ &= P(N(0, 1) > z_\alpha - \sqrt{n}\theta_1) \\ &= 1 - \Phi(z_\alpha - \sqrt{n}\theta_1) \\ &\xrightarrow{n \rightarrow \infty} 1 \end{aligned}$$

If instead, we take $\theta_1 = \frac{h}{\sqrt{n}}$ where h fixed.

$$\text{Power}(h) = 1 - \Phi(z_\alpha - h)$$

This is the **asymptotic local power**.

Example. GLRT for $X_i \sim \text{Uniform}(0, \theta)$

Let $X_1, X_2, \dots, X_n \sim \text{Uniform}(0, \theta)$ i.i.d. We want to test

$$H_0 : \theta = 1 \text{ versus } H_1 : \theta < 1.$$

The likelihood under the uniform model is

$$L(\theta | X_1, \dots, X_n) = \begin{cases} \theta^{-n}, & \text{if } X_{(n)} \leq \theta, \\ 0, & \text{otherwise,} \end{cases}$$

where $X_{(n)} = \max\{X_1, \dots, X_n\}$.

The MLE of θ is therefore

$$\hat{\theta} = X_{(n)}.$$

The generalized likelihood ratio is

$$\Lambda = \frac{L(H_0)}{\max_{\theta} L(\theta)} = \frac{L(\theta = 1)}{L(\theta = \hat{\theta})} = \frac{1^{-n} \mathbf{1}_{\{X_{(n)} \leq 1\}}}{\hat{\theta}^{-n} \mathbf{1}_{\{X_{(n)} \leq \hat{\theta}\}}} = X_{(n)}^n.$$

We reject H_0 for small values of Λ , that is,

$$\Lambda < c_1 \iff X_{(n)} < c_1^{1/n}.$$

Determining c_1 (level α test) We choose c_1 so that the test has size α :

$$P_{\theta=1}(X_{(n)} < c) = \alpha.$$

Under H_0 , since $X_i \sim \text{Uniform}(0, 1)$,

$$P_{\theta=1}(X_{(n)} < c) = P(X_1 < c, \dots, X_n < c) = c^n.$$

Hence,

$$c = \alpha^{1/n}.$$

Therefore, the GLRT of level α rejects H_0 when

$$X_{(n)} < \alpha^{1/n}$$

Power function For $\theta = \theta_0 < 1$,

$$\text{Power}(\theta_0) = P_{\theta_0}(X_{(n)} < \alpha^{1/n}) = P_{\theta_0}(X_1 < \alpha^{1/n}, \dots, X_n < \alpha^{1/n}) = \left(\frac{\alpha^{1/n}}{\theta_0}\right)^n,$$

if $\alpha^{1/n} < \theta_0$, and 1 otherwise.

That is,

$$\text{Power}(\theta_0) = \begin{cases} \left(\frac{\alpha^{1/n}}{\theta_0}\right)^n, & \alpha^{1/n} < \theta_0, \\ 1, & \alpha^{1/n} \geq \theta_0. \end{cases}$$

Local power For local alternatives $\theta_0 = 1 - \frac{h}{n}$, we have

$$\text{Power}(h) = \begin{cases} \frac{\alpha}{(1 - \frac{h}{n})^n}, & \alpha^{1/n} < 1 - \frac{h}{n}, \\ 1, & \alpha^{1/n} \geq 1 - \frac{h}{n}. \end{cases}$$

As $n \rightarrow \infty$,

$$(1 - \frac{h}{n})^{-n} \rightarrow e^h,$$

so that

$$\text{Power}(h) = \begin{cases} \alpha e^h, & \text{if } \alpha e^h < 1, \\ 1, & \text{if } \alpha e^h \geq 1. \end{cases}$$

20 Lecture 22

20.1 Multicollinearity

Definition 20.1. Multicollinearity: This occurs when one (or more) of the predictors is nearly a linear combination of some of the others.

This leads to higher variance of the estimates of the regression coefficients. The coefficients may have large standard errors and become statistically insignificant, even when the overall F-statistic is significant.

20.2 Variance Inflation Factor (VIF)

Definition 20.2. Consider the regression of the variable X_j on the remaining predictors $\{X_k : k \neq j\}$. Let R_j^2 denote the coefficient of determination in this regression.

The **variance inflation factor** of X_j is defined as

$$\text{VIF}_j = \frac{1}{1 - R_j^2}.$$

The VIF measures how much the variance of the LS estimate is inflated due to collinearity.

20.3 Interpretation of VIF

Consider the simple (univariate) regression of Y on X_j :

$$Y_i = \alpha_0 + \alpha_j X_{ji} + \varepsilon_i.$$

Then

$$\mathbb{V}(\hat{\alpha}_j) = \frac{\sigma^2}{s_j^2},$$

where

$$s_j^2 = \sum_{i=1}^n (X_{ji} - \bar{X}_j)^2.$$

Result 20.3. In the full regression model

$$Y = X\beta + \varepsilon,$$

using the formula for the inverse of a partitioned matrix, one can show

$$\mathbb{V}(\hat{\beta}_j) = \frac{\sigma^2}{1 - R_j^2} \cdot \frac{1}{s_j^2}.$$

Thus the variance of $\hat{\beta}_j$ is inflated relative to the univariate case by the factor

$$\text{VIF}_j = \frac{1}{1 - R_j^2}.$$

20.4 Methods to Handle Multicollinearity

- Remove some variables (to reduce collinearity in the remaining ones).
- Use shrinkage methods such as ridge regression.

20.5 Ridge Regression

Consider the multiple linear regression model

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_p X_{pi} + \varepsilon_i,$$

for $1 \leq i \leq n$, where p may be larger than n .

Given a regularization parameter $\lambda > 0$, the ridge regression estimator is

$$\hat{\beta}(\lambda) = \arg \min_{\beta} \left\{ \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji} \right)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right\}.$$

Remarks.

1. Larger λ means more shrinkage toward zero.
2. If $\lambda = 0$, we recover the least squares estimator.
3. The intercept term is **not** penalized.

20.6 Centering and Scaling

Typically the predictors are scaled to mean 0 and variance 1:

$$\bar{X}_j = \frac{1}{n} \sum_{i=1}^n X_{ji}, \quad s_j^2 = \frac{1}{n-1} \sum_{i=1}^n (X_{ji} - \bar{X}_j)^2,$$

and the standardized version is

$$\tilde{X}_{ji} = \frac{X_{ji} - \bar{X}_j}{s_j}.$$

This removes the intercept and makes ridge coefficients equivariant under scaling transformations

$$X_{ji} \mapsto a + bX_{ji}.$$

20.7 Constrained Form of Ridge Regression

The equivalent constrained optimization problem is

$$\hat{\beta}(\lambda) = \arg \min_{\beta} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji} \right)^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq t.$$

20.8 Matrix Form

Assume Y is centered and X is scaled so that X has p columns. Then the ridge objective is

$$RSS(\lambda) = \|Y - X\beta\|^2 + \lambda\|\beta\|^2.$$

Then

$$\hat{\beta}(\lambda) = \arg \min_{\beta} RSS(\lambda).$$

Result 20.4. The ridge regression estimator is

$$\hat{\beta}(\lambda) = (X^\top X + \lambda I_p)^{-1} X^\top Y,$$

where I_p is the $p \times p$ identity matrix.

1. $\hat{\beta}(\lambda)$ is a linear function of Y .
2. $X^\top X + \lambda I_p$ is invertible even when $X^\top X$ is not.

Proof. Taking the gradient of the objective:

$$\nabla_{\beta} RSS(\lambda) = -2X^\top Y + 2X^\top X\beta + 2\lambda\beta = 0.$$

Thus

$$(X^\top X + \lambda I_p)\hat{\beta}(\lambda) = X^\top Y,$$

and hence

$$\hat{\beta}(\lambda) = (X^\top X + \lambda I_p)^{-1} X^\top Y.$$

✓

21 Lecture 23

Recall that the ridge regression estimator is

$$\hat{\beta}(\lambda) = (X^\top X + \lambda I_p)^{-1} X^\top Y.$$

21.1 Properties of the Ridge Regression Estimator

Assume $\varepsilon \sim N_n(0, \sigma^2 I)$.

Result 21.1.

$$\mathbb{E}[\hat{\beta}(\lambda)] = (X^\top X + \lambda I)^{-1} X^\top \mathbb{E}[Y] = (X^\top X + \lambda I)^{-1} X^\top X \beta = \beta - \lambda (X^\top X + \lambda I)^{-1} \beta.$$

Thus, $\hat{\beta}(\lambda)$ is **biased**.

Result 21.2.

$$\mathbb{V}(\hat{\beta}(\lambda)) = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1}.$$

Result 21.3. It can be shown that

$$\mathbb{V}(\hat{\beta}_{LS}) - \mathbb{V}(\hat{\beta}(\lambda))$$

is a nonnegative definite matrix. Hence ridge regression has **smaller variance** than least squares.

Result 21.4. $\hat{\beta}(\lambda)$ is multivariate normal with mean and variance as above.

21.2 Mean-Square Error for a Multivariate Estimator

Suppose $\hat{\theta}$ estimates a p -dimensional parameter θ .

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^\top (\hat{\theta} - \theta)].$$

Expanding,

$$\text{MSE}(\hat{\theta}) = \sum_{i=1}^p \mathbb{V}(\hat{\theta}_i) + \|\text{Bias}(\hat{\theta})\|^2 = \text{tr}(\mathbb{V}(\hat{\theta})) + \|\text{Bias}(\hat{\theta})\|^2.$$

21.3 Degrees of Freedom in Ridge Regression

Define fitted values

$$\hat{Y}(\lambda) = X \hat{\beta}(\lambda) = X (X^\top X + \lambda I)^{-1} X^\top Y = H_\lambda Y,$$

where

$$H_\lambda = X (X^\top X + \lambda I)^{-1} X^\top.$$

For $\lambda = 0$, $H_0 = P_X$, the projection matrix onto $\text{col}(X)$.

Definition 21.5. The **effective degrees of freedom** of ridge regression is

$$df(\lambda) = \text{tr}(H_\lambda).$$

Result 21.6. If $d_1, \dots, d_p$ are the singular values of X , then

$$df(\lambda) = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda}.$$

Remarks.

1. If the eigenvalues of $X^\top X$ are $\kappa_1, \dots, \kappa_p$, then $d_j = \sqrt{\kappa_j}$.
2. $df(0) = p$, as in least squares.
3. $df(\lambda) \leq df(0)$ for all $\lambda > 0$.

21.4 Selecting λ : Generalized Cross-Validation (GCV)

Define

$$\text{GCV}(\lambda) = \frac{1}{n} \sum_{i=1}^n \left(\frac{Y_i - \hat{Y}_i}{1 - \frac{df(\lambda)}{n}} \right)^2.$$

Interpretation: For each i , fit the model leaving out observation i and denote the predicted value by $\hat{Y}_{-i}(X_i)$. A measure of predictive error is

$$\frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_{-i}(X_i))^2.$$

It can be shown this equals approximately the expression above.

21.5 Alternate Constrained Form of Ridge Regression

$$\hat{\beta}(\lambda) = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji})^2 \quad \text{subject to} \quad \sum_{j=1}^p \beta_j^2 \leq K.$$

21.6 Best Subset Selection

Choose the “best” model with at most $K \leq p$ variables:

$$\hat{\beta}(K) = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji})^2 \quad \text{such that} \quad \sum_{j=1}^p 1_{\{\beta_j \neq 0\}} \leq K.$$

This is a **nonconvex** optimization problem.

21.7 LASSO (Least Absolute Shrinkage and Selection Operator)

$$\hat{\beta}(K) = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji})^2 \quad \text{such that} \quad \sum_{j=1}^p |\beta_j| \leq K.$$

Equivalent Lagrangian form:

$$\hat{\beta}(\lambda) = \arg \min_{\beta} \frac{1}{2} \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji})^2 + \lambda \sum_{j=1}^p |\beta_j|.$$

Remarks.

1. LASSO uses an ℓ_1 penalty. The solution is nonlinear in Y —no closed form exists.
2. The problem is convex; efficient solvers exist.
3. If $K > \sum_j |\hat{\beta}_{LS,j}|$, then LASSO reduces to LS.
4. The ℓ_1 penalty induces **sparsity** in $\hat{\beta}$.

22 Lecture 24

22.1 Best Subset Selection

Choose the “best” model with at most $K \leq p$ variables:

$$\hat{\beta}(K) = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji})^2 \quad \text{such that} \quad \sum_{j=1}^p 1_{\{\beta_j \neq 0\}} \leq K.$$

This is a **non-convex** optimization problem and is computationally difficult.

22.2 LASSO (Least Absolute Shrinkage and Selection Operator)

$$\hat{\beta}(K) = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji})^2 \quad \text{such that} \quad \sum_{j=1}^p |\beta_j| \leq K.$$

22.2.1 Lagrangian Form

$$\hat{\beta}(\lambda) = \arg \min_{\beta} \left\{ \frac{1}{2} \sum_{i=1}^n (Y_i - \beta_0 - \sum_{j=1}^p \beta_j X_{ji})^2 + \lambda \sum_{j=1}^p |\beta_j| \right\}.$$

Remarks.

1. The ℓ_1 penalty induces sparsity.
2. The solution is **nonlinear** in Y ; no closed form exists.
3. The problem is convex, so very efficient numerical solvers exist.
4. If $K > \sum_j |\hat{\beta}_j^{LS}|$, then the constraint is inactive and

$$\hat{\beta}_{LASSO}(K) = \hat{\beta}_{LS}.$$

22.3 Closed Form Shrinkage When $X = I$

Assume $X = I_{p \times p}$ so $p = n$ and the model is:

$$y_i = \beta_i + \varepsilon_i.$$

Ridge Regression.

$$\hat{\beta}_i(\lambda) = \frac{y_i}{1 + \lambda}.$$

LASSO. Solve

$$\hat{\beta}_i(\lambda) = \arg \min_{\beta_i} \left\{ \frac{1}{2} (y_i - \beta_i)^2 + \lambda |\beta_i| \right\}.$$

By calculus:

$$\hat{\beta}_i(\lambda) = \begin{cases} y_i - \lambda, & y_i > \lambda, \\ 0, & |y_i| \leq \lambda, \\ y_i + \lambda, & y_i < -\lambda. \end{cases}$$

This is the **soft-thresholding estimator**.

Best Subset Selection.

$$\hat{\beta}_i(K) = \begin{cases} y_i, & |y_i| > \lambda, \\ 0, & \text{otherwise.} \end{cases}$$

This is the **hard-thresholding estimator**.

22.4 Selecting λ in LASSO: K -Fold Cross-Validation

Fix $\lambda > 0$.

1. Split the data into K roughly equal parts.
2. For fold i , fit the model on the remaining $K - 1$ parts and compute prediction error on fold i .
3. Repeat for all $i = 1, \dots, K$ and sum the prediction errors.
4. Choose the value of λ that minimizes the cross-validated error. Alternatively, choose the largest λ within one standard error of the minimum.

Formally, let $K(i)$ indicate the fold assignment of observation i . Then:

$$CV(\lambda) = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_{-K(i)}(X_i))^2,$$

where $\hat{Y}_{-K(i)}(X_i)$ is the prediction from a model trained **excluding** fold $K(i)$.

22.5 Elastic Net

If predictors are highly correlated, LASSO tends to select only one and ignore the rest. Elastic Net uses a mixture of ℓ_1 and ℓ_2 penalties:

$$\hat{\beta}(\lambda_1, \lambda_2) = \arg \min_{\beta} \{ \|Y - X\beta\|^2 + \lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2 \}.$$

This combines ridge and LASSO to stabilize selection under correlation.

22.6 Logistic Regression

Assume $Y_i \sim \text{Bern}(p_i)$ independently, and model the log-odds:

$$\log \left(\frac{p_i}{1 - p_i} \right) = x_i^\top \beta.$$

Equivalently:

$$p_i = P(Y_i = 1 | x_i) = \frac{e^{x_i^\top \beta}}{1 + e^{x_i^\top \beta}}.$$

22.7 Likelihood and Estimation

The likelihood is:

$$L(\beta) = \prod_{i=1}^n p_i^{Y_i} (1 - p_i)^{1 - Y_i}.$$

Log-likelihood:

$$\ell(\beta) = \sum_{i=1}^n \left\{ Y_i x_i^\top \beta - \log(1 + e^{x_i^\top \beta}) \right\}.$$

Score equations:

$$0 = \frac{\partial \ell}{\partial \beta_j} = \sum_{i=1}^n x_{ji} (Y_i - p_i),$$

with $p_i = \frac{e^{x_i^\top \beta}}{1 + e^{x_i^\top \beta}}$.

These are solved numerically (e.g., Newton–Raphson, IRLS).

For a new data point x_{new} :

$$\hat{p} = \frac{e^{x_{\text{new}}^\top \hat{\beta}}}{1 + e^{x_{\text{new}}^\top \hat{\beta}}}.$$

22.8 GLRT for Testing a Coefficient

Test:

$$H_0 : \beta_p = 0 \quad \text{vs} \quad H_1 : \beta_p \neq 0.$$

Let $\hat{\beta}$ be the MLE under the full model, and $\hat{\beta}_0$ the MLE under H_0 . Reject when

$$-2 \log \Lambda = 2 \left[\ell(\hat{\beta}) - \ell(\hat{\beta}_0) \right] > \chi_{1, \alpha}^2.$$

22.9 Confidence Interval for β_j

The Fisher information is:

$$I(\beta) = X^\top W X,$$

where

$$W = \text{diag} \left(\frac{e^{x_i^\top \beta}}{(1 + e^{x_i^\top \beta})^2} \right).$$

Asymptotically:

$$\hat{\beta} \approx N_{p+1}(\beta, (X^\top W X)^{-1}).$$

Let $\hat{W} = W(\hat{\beta})$. Then an approximate $100(1 - \alpha)\%$ confidence interval for β_j is:

$$\hat{\beta}_j \pm z_{\alpha/2} \sqrt{(X^\top \hat{W} X)^{-1}_{jj}}.$$

23 Appendix

Property	Bernoulli	Normal	Laplace	Poisson	Uniform $[a, b]$
PDF / PMF	$f(x) = p^x(1-p)^{1-x}, x \in \{0, 1\}$	$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$	$f(x) = \frac{1}{2b} e^{-\frac{ x-\mu }{b}}$	$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}$	$f(x) = \frac{1}{b-a}, x \in [a, b]$
Variance	$p(1-p)$	σ^2	$2b^2$	λ	$\frac{(b-a)^2}{12}$
Mean	p	μ	μ	λ	$\frac{a+b}{2}$
$\mathbb{E}[X^2]$	p	$\mu^2 + \sigma^2$	$\mu^2 + 2b^2$	$\lambda + \lambda^2$	$\frac{a^2+ab+b^2}{3}$
$\mathbb{E}[X^4]$	p	$\mu^4 + 6\mu^2\sigma^2 + 3\sigma^4$	$\mu^4 + 12b^2\mu^2 + 24b^4$	$\lambda^4 + 6\lambda^3 + 7\lambda^2 + \lambda$	$\frac{a^4+a^3b+a^2b^2+ab^3+b^4}{5}$
MLE estimates	$\hat{p} = \frac{1}{n} \sum x_i$	$\hat{\mu} = \frac{1}{n} \sum x_i, \hat{\sigma}^2 = \frac{1}{n} \sum (x_i - \hat{\mu})^2$	$\hat{\mu} = \text{median}(x_i), \hat{b} = \frac{1}{n} \sum x_i - \hat{\mu} $	$\hat{\lambda} = \frac{1}{n} \sum x_i$	$\hat{a} = \min x_i, \hat{b} = \max x_i$

Table 1: Common distributions: densities, moments, and MLEs.

Test Type	Setting / Assumptions	Test Statistic	Rejection Region (Level α)	Remarks / Examples
Most Powerful (MP)	Simple vs. Simple $H_0 : \theta = \theta_0$ vs $H_1 : \theta = \theta_1$	Likelihood Ratio: $\Lambda(x) = \frac{f_{\theta_0}(x)}{f_{\theta_1}(x)}$	Reject H_0 when $\Lambda(x) < c_\alpha$ (c_α chosen so $P_{H_0}(\Lambda < c_\alpha) = \alpha$)	Guaranteed most powerful by Neyman-Pearson Lemma . Example: Normal mean test with known σ^2 .
Uniformly Most Powerful (UMP)	Simple vs. One-Sided Composite $H_0 : \theta = \theta_0$ vs $H_1 : \theta > \theta_0$ (or $< \theta_0$) Family with Mono-tone Likelihood Ratio (MLR)	Statistic $T(X)$ s.t. $\frac{f_{\theta_1}(x)}{f_{\theta_0}(x)}$ increases in T	Reject H_0 for large $T(X)$ if $H_1 : \theta > \theta_0$ (or small if $H_1 : \theta < \theta_0$)	UMP test exists if MLR holds. Example: $X_i \sim N(\mu, \sigma^2)$ known σ^2 , reject for large $\bar{X}$. Power increases with μ .
No UMP (Two-Sided)	Simple vs. Two-Sided Composite $H_0 : \theta = \theta_0$ vs $H_1 : \theta \neq \theta_0$	Varies by test (see GLRT)	Reject when $ T(X) > c_\alpha$ where $T(X)$ symmetric about 0	No single test uniformly best. Use GLRT or split-tail test. Example: two-sided z or t test for mean.
Generalized Likelihood Ratio Test (GLRT)	Composite vs. Composite $H_0 : \theta \in \Omega_0$ vs $H_1 : \theta \in \Omega \setminus \Omega_0$	$\Lambda = \frac{\max_{\theta \in \Omega_0} L(\theta)}{\max_{\theta \in \Omega} L(\theta)}$	Reject H_0 for small Λ or large $-2 \log \Lambda$	General-purpose test when MP/UMP doesn't exist. Asymptotic result: $-2 \log \Lambda \xrightarrow{d} \chi_d^2$, $d = \dim(\Omega) - \dim(\Omega_0)$. Example: testing $H_0 : \mu = \mu_0$ in $N(\mu, \sigma^2)$ (unknown σ^2) $\Rightarrow$ t -test.
Wald Test	Large-Sample Asymptotic $H_0 : \theta = \theta_0$	$W = \frac{(\hat{\theta} - \theta_0)^2}{\widehat{\text{Var}}(\hat{\theta})}$	Reject H_0 if $W > \chi_{1,\alpha}^2$	Based on MLE asymptotic normality. Equivalent to square of z -test. Used in GLM / MLE frameworks.
Score (Lagrange Multiplier) Test	Large-Sample Asymptotic $H_0 : \theta = \theta_0$	$S = \frac{[s(\theta_0)]^2}{I(\theta_0)}, \quad s(\theta) = \partial_\theta \log L(\theta)$	Reject H_0 if $S > \chi_{1,\alpha}^2$	Uses only score at null hypothesis. Convenient when fitting under H_1 is hard. Asymptotically equivalent to Wald and LRT.
Likelihood Ratio (Asymptotic form)	General large-sample case	$-2 \log \Lambda = 2[\log L(\hat{\theta}) - \log L(\hat{\theta}_0)]$	Reject if $-2 \log \Lambda > \chi_{d,\alpha}^2$	Asymptotically UMP invariant. Equivalent to Wald/Score as $n \rightarrow \infty$.

Table 2: Unified summary of MP, UMP, GLRT, Wald, and Score tests under different assumptions.